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A CENTRALIZED SCHOLARSHIP MANAGEMENT AND TRACKING SYSTEM FOR BULACAN STATE UNIVERSITY – BUSTOS CAMPUS

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CHAPTER I:

THE PROBLEM AND ITS BACKGROUND

Introduction

Scholarship programs play an important role in supporting students' access to higher education by providing financial assistance to qualified and deserving learners. In universities, scholarship programs help reduce financial barriers and provide students with opportunities to continue their studies. However, as the number of applicants, scholarship programs, requirements, and grantors increases, managing scholarship records becomes more difficult when processes are handled manually or through fragmented systems.

Manual scholarship processing often results in delays, data errors, incomplete records, and communication problems between applicants and administrators. Jibrin et al. (2016) developed an e-scholarship system after identifying that paper-based scholarship application processing was time-consuming and prone to errors, especially during submission, screening, evaluation, and communication of application concerns. Similarly, Ablen et al. (2025) explained that manual scholarship administration may cause delayed processing, inaccurate record-keeping, and poor communication among stakeholders. These issues show the need for a centralized digital platform that can support scholarship application, validation, tracking, and communication.

At Bulacan State University – Bustos Campus, scholarship-related processes are handled by the Institutional Student Programs and Services (ISPS). Although scholarship programs are available to students, the current process still involves manual or fragmented procedures such as scholarship posting, application submission, document verification, approval monitoring, and student notification.

These limitations make it difficult to track application status, monitor active scholarships, verify student records, and ensure that scholarship data are accurate and updated.

One major concern in scholarship management is the possibility of duplicate scholarship allocation. Widyastuti and Prasandha (2025) noted that lack of data integration in scholarship administration may result in delays, misallocation, and potential double funding. They also emphasized that centralized digital systems can improve efficiency, control, transparency, and audit trails. In relation to this, the developed system includes a pre-approval validation mechanism that checks whether a student already has an active scholarship before approval. This ensures that duplication is prevented before a scholarship is granted instead of being detected after approval.

To address these concerns, this study developed a centralized web-based scholarship management and tracking system for Bulacan State University – Bustos Campus. The system supports scholarship posting, student application, document submission, scholarship recommendation, pre-approval validation, approval, monitoring, renewal, notification, and reporting. A web-based scholarship management system is appropriate because it can provide centralized access to scholarship information, support application submission, improve information dissemination, and promote transparency in the scholarship process (Al-Ayyubi & Maulana, 2023).

Background of the Study

Scholarship programs are designed to support students in pursuing higher education by providing financial assistance and academic opportunities. These programs are commonly managed by educational institutions, government agencies, private organizations, and other scholarship providers. Effective scholarship management requires accurate record-keeping, organized application processing, systematic evaluation, timely communication, and proper monitoring of scholars.

At Bulacan State University – Bustos Campus, scholarship-related processes are handled by the Institutional Student Programs and Services (ISPS). The office is responsible for managing scholarship applications, renewals, endorsements, and monitoring activities. Although scholarship programs are available to students, some processes still rely on manual or fragmented procedures, including application submission, document verification, record tracking, approval monitoring, and student notification. This setup may result in delays, data inconsistencies, and increased administrative workload.

Manual and fragmented scholarship processes can also affect the accuracy and transparency of scholarship records. Jibrin et al. (2016) explained that paper-based scholarship application processing is time-consuming and prone to errors, especially when applicants submit forms manually and administrators process applications through manual screening and evaluation. Similarly, Ablen et al. (2025) noted that manual scholarship administration may lead to delayed processing, inaccurate record-keeping, and poor communication among stakeholders.

One of the major concerns in scholarship management is ensuring that only one active scholarship is granted to a student at a given time. Without a centralized validation process, scholarship duplication may occur when records are stored separately or when scholarship information comes from different sources. Widyastuti and Prasandha (2025) stated that lack of data integration in scholarship administration may lead to delays, misallocation, and potential double funding. They also emphasized that digital and centralized systems can improve efficiency, transparency, data control, and accountability.

In response to these concerns, the study developed a centralized web-based scholarship management and tracking system for Bulacan State University – Bustos Campus. The system provides a unified platform where students can view scholarship opportunities, submit applications and required

documents, and track their application status. Administrators and grantors can manage scholarship postings, evaluate applicants, process approvals, monitor scholars, and generate reports.

The developed system also includes pre-approval validation to help prevent multiple active scholarships, as well as scholarship recommendation, monitoring, renewal tracking, notifications, and reporting features. Through these functions, the system is designed to improve scholarship administration by reducing manual workload, improving data accuracy, supporting timely communication, and strengthening transparency and accountability in scholarship management.

Objectives of the Study

General Objective

To design, develop, and evaluate a centralized scholarship management and tracking system that streamlines application, approval, and monitoring processes while preventing duplicate scholarship allocation.

Specific Objectives

Specifically, this study aims:

1. To develop a centralized scholarship management system that enables students to apply and submit required documents, while allowing grantors and administrators to manage scholarship postings, evaluate applicants, and process approvals efficiently within a unified platform.
2. To develop a system that validates student records, prevents multiple active scholarships, and enforces duplication prevention across both internal and externally sourced scholarship records.

3. To develop a system that provides scholarship recommendations, manages the scholarship lifecycle (application, approval, active status, renewal, and completion), and monitors student compliance and academic performance.
4. To evaluate the acceptability of the system using the Technology Acceptance Model (TAM) in terms of:
    a) Perceived Usefulness
    b) Perceived Ease of Use
    c) Behavioral Intention
5. To evaluate the quality of the developed system based on the ISO/IEC 25010 (2023) software quality model in terms of:
    a) Functional Suitability
    b) Performance Efficiency
    c) Compatibility
    d) Usability
    e) Reliability
    f) Security
    g) Maintainability
    h) Portability

Conceptual Framework

This study is anchored on the Input–Process–Output (IPO) model, which illustrates how the developed **centralized scholarship management and tracking system** transforms inputs into meaningful outputs to support effective scholarship administration, monitoring, and record management at Bulacan State University – Bustos Campus.

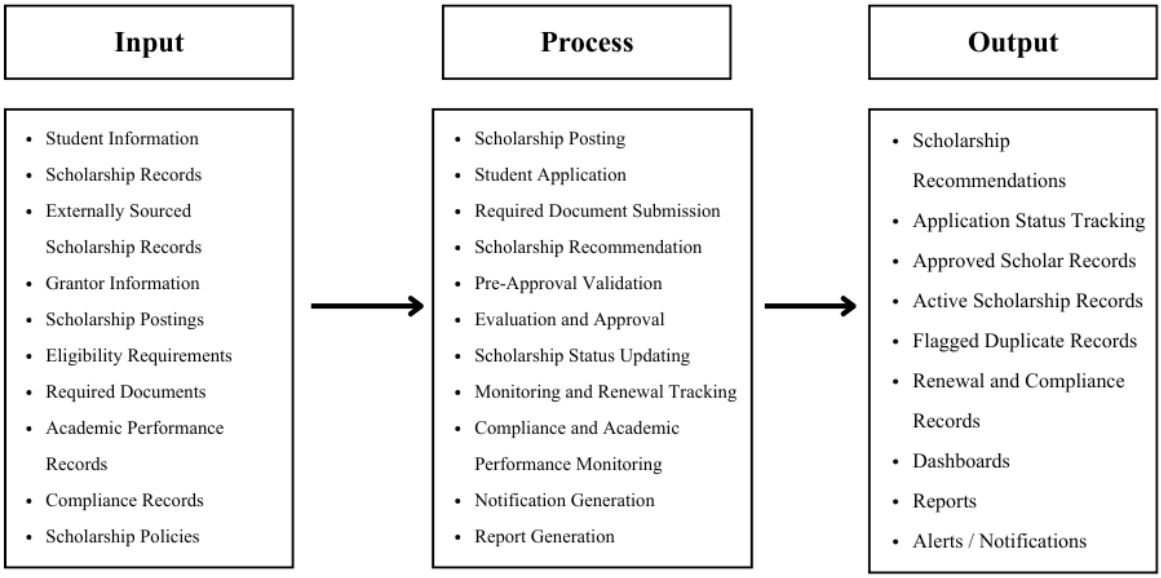


Figure 1 presents the conceptual framework of the study using the Input–Process–Output (IPO) model. The model illustrates how the developed centralized scholarship management and tracking system processes student, scholarship, grantor, requirement, and policy data to produce organized scholarship records, recommendations, monitoring reports, notifications, and tracking outputs.

The **input stage** includes essential data such as student information, scholarship records, externally sourced scholarship records, grantor information, scholarship postings, eligibility requirements, required documents, academic performance records, compliance records, and

scholarship policies. Student information includes personal, academic, and financial data used to identify and verify applicants. Scholarship records include existing internal and externally sourced scholarship data used for monitoring and duplication prevention. Grantor information refers to scholarship providers who manage scholarship programs within the system. Scholarship policies define eligibility rules, renewal conditions, compliance requirements, and the enforcement of one active scholarship per student.

The **process stage** represents the system operations that transform input data into organized and actionable information. The process begins with scholarship posting, where grantors or administrators create and publish available scholarship programs, including eligibility requirements, required documents, deadlines, and available slots. This is followed by student application, where students apply and submit required documents through the system. The system then provides scholarship recommendations by comparing student records with scholarship eligibility requirements.

After application submission, the system performs pre-approval validation to check whether the student already has an active scholarship. This validation helps prevent duplicate scholarship allocation before approval. Grantors and administrators then evaluate applications and approve, reject, or request corrections from applicants. Once a student is approved, the system updates the scholarship record, marks the student's scholarship status as active, and prevents the student from being approved for another active scholarship. The system also supports monitoring, renewal tracking, compliance checking, academic performance monitoring, notification generation, and report generation throughout the scholarship lifecycle.

The **output stage** represents the final results generated by the system. These outputs include scholarship recommendations, real-time application status tracking, approved scholar records, active scholarship records, flagged duplicate records, renewal and compliance records, dashboards, reports, and alerts or notifications. These outputs improve transparency, efficiency, accountability, and decision-making in scholarship management at Bulacan State University – Bustos Campus.

Scope and Limitations of the Study

Scope of the Study

This study focuses on the design, development, and evaluation of a centralized scholarship management and tracking system for Bulacan State University – Bustos Campus. The system supports the scholarship process, including scholarship posting, student application, document submission, evaluation, approval, monitoring, renewal, and completion.

The system allows students to view available scholarship opportunities, apply for scholarships, submit required documents, and track the status of their applications. It also allows grantors and administrators to manage scholarship postings, review submitted applications, evaluate applicants, process approvals, monitor scholars, and generate reports.

The system manages student scholarship records and supports pre-approval validation to help prevent multiple active scholarships from being granted to the same student. It includes both internal scholarship records and externally sourced scholarship records that are encoded into the system for monitoring and duplication prevention.

The system also provides scholarship recommendations based on student records and scholarship eligibility requirements. It supports scholarship lifecycle management by tracking application, approval, active status, renewal, and completion. It also monitors student compliance and academic performance as part of scholarship monitoring.

The system includes notification and reporting features to inform students, grantors, and administrators about scholarship postings, application status, document requirements, deadlines, renewal updates, and monitoring records. Notifications may be delivered through in-system alerts and email notifications.

The study involves students, scholarship grantors, and ISPS coordinators or administrators under the Institutional Student Programs and Services office of Bulacan State University – Bustos Campus.

Limitations of the Study

This study is limited to the scholarship management processes of Bulacan State University – Bustos Campus and does not cover other campuses of Bulacan State University. Only scholarship programs administered, recognized, or encoded within the campus system are included.

The system does not include financial disbursement, fund release, payroll, liquidation, or accounting-related processes. It focuses only on scholarship posting, application, document submission, validation, evaluation, approval, monitoring, renewal, completion, notification, and reporting.

The system does not provide full real-time system-to-system integration with external scholarship providers such as DOST, UNIFAST, other government agencies, or private scholarship databases. However, externally sourced scholarship records may be manually encoded into the system so they can be included in scholarship monitoring and duplication prevention. Future API-based integration may be considered when access, data-sharing agreements, and security permissions become available.

The accuracy of the system depends on the correctness and completeness of the data provided by students, grantors, and administrators. Incomplete, outdated, or incorrectly encoded records may affect the accuracy of scholarship recommendations, validation results, monitoring records, and reports.

The system supports the monitoring of student compliance and academic performance, but final decisions regarding approval, renewal, cancellation, or termination of scholarships remain under the authority of the assigned administrators, grantors, or ISPS coordinators.

The system is developed as a web-based platform and does not include a native mobile application. However, it may be accessed through supported web browsers using compatible devices.

Significance of the Study

Students

The developed system provides students with a centralized platform where they can view available scholarship opportunities, receive suitable scholarship recommendations, submit required documents, and track their application status. This can reduce confusion, missed deadlines, and delays in scholarship application processing. Web-based scholarship systems can improve access to scholarship information and support faster dissemination of updates to students (Al-Ayyubi & Maulana, 2023).

Scholarship Grantors

The system helps scholarship grantors manage scholarship postings, eligibility requirements, application records, evaluation results, and approved scholars. It also helps them monitor scholarship slots and generate reports. By using a centralized system, grantors can reduce manual workload and improve coordination with administrators and students.

ISPS Coordinators and Administrators

The developed system assists ISPS coordinators and administrators in managing scholarship records, validating applications, monitoring active scholars, preventing duplicate scholarship allocation, and generating reports. Centralized scholarship systems can improve accountability by strengthening transparency, data control, and audit trails (Widyastuti & Prasandha, 2025).

Bulacan State University – Bustos Campus

The system supports improved transparency, efficiency, and accountability in scholarship management. The system supports better decision-making through centralized reports, monitoring records, and organized scholarship data. It can also help the campus enforce scholarship policies, such as preventing multiple active scholarships from being granted to the same student.

System Developers and IT Personnel

The study may serve as a reference for system developers and IT personnel designing or improving scholarship management systems. It provides an example of how scholarship posting, application, recommendation, duplication prevention, monitoring, and reporting can be centralized into a single web-based platform.

Future Researchers

Future researchers may use this study as a reference for improving scholarship management systems, recommendation systems, duplication prevention mechanisms, and educational information systems. They may also extend the system by adding full external database integration, mobile application support, advanced analytics, or automated API-based verification with external scholarship providers.

CHAPTER II:

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the literature, technical concepts, related studies, and related systems that support the development of the developed centralized scholarship management and tracking system for Bulacan State University – Bustos Campus. It discusses scholarship management systems, web-based platforms, centralized data management, scholarship recommendation, duplication prevention, and system evaluation models. The reviewed materials serve as the foundation for understanding the need for a centralized system that can support scholarship application, validation, monitoring, and reporting.

Technical Background

Web-Based Information System

A web-based information system is an application that can be accessed through a web browser and used by different users through the internet or a local network. In scholarship management, a web-based system allows students, administrators, and grantors to access scholarship information, submit required documents, monitor applications, and receive updates without requiring installation on individual devices.

Al-Ayyubi and Maulana (2023) explained that web-based scholarship systems can improve access to scholarship information, facilitate the rapid dissemination of updates, and provide

features such as centralized database management, application submission, document management, search functionality, security, and notifications. These features are relevant to the developed system because it aims to provide one platform for scholarship posting, application, monitoring, and reporting.

In the context of this study, the developed system is a web-based platform to allow students to view scholarship opportunities, submit required documents, and track application status. Administrators and grantors can also manage scholarship postings, validate applications, monitor scholars, and generate reports through the same system.

Centralized Scholarship Management

Centralized scholarship management uses a single system or database to manage scholarship-related data instead of storing records separately across offices, files, or disconnected platforms. A centralized system improves the organization of student records, scholarship postings, application data, monitoring records, and reports.

Widyastuti and Prasandha (2025) emphasized that lack of data integration in scholarship administration may result in delays, misallocation, and potential double funding. They also stated that centralized digital systems can improve efficiency, control, transparency, and audit trails. This supports the need for the developed system to centralize scholarship records and help prevent duplicate scholarship allocation.

For Bulacan State University – Bustos Campus, centralization is important because scholarship information may come from different sources, including internal scholarship programs and

externally sourced scholarship records encoded into the system. By consolidating these records, the system can improve data accuracy, support better monitoring, and help administrators identify possible duplication before approval.

Role-Based Access Control

Role-Based Access Control, or RBAC, is a security approach in which system access is granted based on user roles. In a scholarship management system, different users have different responsibilities. Students may apply for scholarships and track their application status, grantors may post scholarships and evaluate applicants, and administrators may manage users, validate records, monitor scholars, and generate reports.

Amer (2025) included role-based access, dashboards, reports, and scholarship management features in a smart scholarship management system to support transparency and accountability. In this study, RBAC is used to ensure that each user can only access the functions and records appropriate to their role. This help protect student data and maintain accountability in scholarship processing.

Scholarship Recommendation

Scholarship recommendation refers to the process of identifying suitable scholarship opportunities based on student records and scholarship eligibility requirements. This feature is important because students may have difficulty determining which scholarships match their qualifications, academic standing, financial background, or other eligibility factors.

In the developed system, scholarship recommendations are based on student attributes such as general weighted average, family income, year level, course or program, achievements, residency, and other scholarship-related requirements. These data help the system identify scholarship programs that match the student's profile and eligibility.

Decision Tree algorithms may be used for scholarship recommendation because they classify data through clear decision rules. For example, a student with high academic performance and low family income may be recommended for an academic or financial assistance scholarship depending on the scholarship requirements. C4.5 and CART are established decision tree approaches used for classification tasks (Quinlan, 1993; Breiman et al., 1984).

Daluyon and Bilog (2025) developed a Scholarship Management Information System using machine learning models and found that the Decision Tree achieved the highest accuracy among the tested classifiers. Their study also used student profile data, data preprocessing, and classifier evaluation for scholarship recommendation, prescription, and prioritization. This supports the use of a Decision Tree approach as a possible method for the recommendation feature of the developed system.

The basic recommendation process involves collecting student data, preparing the dataset, comparing student attributes with scholarship eligibility criteria, and classifying students into appropriate scholarship categories. The result helps students identify scholarships that match their qualifications and help administrators support a more systematic recommendation process.

Duplication Prevention

Duplication prevention is one of the important features of the developed system. The system validates student records before approval. This helps prevents a student from receiving more than one active scholarship at the same time.

Record Linkage is a method used to determine whether records from different files refer to the same person, object, or event. Fellegi and Sunter (1969) introduced a theoretical framework for identifying matching records across files. In the developed system, this concept is applied to compare student records from internal scholarship records and externally sourced scholarship records encoded into the system.

The duplication prevention process may begin with exact matching. The system first checks whether the student identification number already exists in the list of active scholars. If the student identification number matches an active scholarship record, the system prevents approval or flags the record for administrator review.

If the student identification number is missing, inconsistent, or different across records, the system may use fuzzy matching. Levenshtein Distance measures the number of edits needed to transform one string into another, such as insertions, deletions, or substitutions (Levenshtein, 1966). In this study, it may be used to compare student names and other text-based fields that may have spelling variations.

The system may compute a similarity score using fields such as student ID, full name, birthdate, school, course, address, and contact number. If the similarity score reaches the

defined threshold, the system may flag the record as a possible duplicate and prevent approval until the administrator reviews the result. This approach strengthens duplication prevention by using both exact and similarity-based record checking rather than relying only on a simple manual review.

Technology Acceptance Model

The Technology Acceptance Model, or TAM, is used to evaluate whether users are likely to accept and use a system. It focuses on factors such as perceived usefulness and perceived ease of use. Davis (1989) explained that perceived usefulness and perceived ease of use are important determinants of user acceptance of information technology.

In this study, TAM was used to evaluate the acceptability of the developed system in terms of perceived usefulness, perceived ease of use, and behavioral intention. This helps determine whether students, grantors, and administrators find the system useful, easy to use, and likely to be adopted in scholarship management processes.

ISO/IEC 25010 Software Quality Model

ISO/IEC 25010 is a software quality model used to evaluate software systems. The standard provides quality characteristics that can be used to assess whether a system meets functional and operational requirements. In this study, ISO/IEC 25010 (2023) was used to evaluate the developed system with respect to functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Daluyon and Bilog (2025) evaluated their Scholarship Management Information System using ISO/IEC 25010 and obtained a "Highly Acceptable" rating. Yan (2024) also evaluated an online students' scholarship evaluation system using ISO 25010 and found that the system complied to a very great extent in several software quality characteristics. These studies support the use of ISO/IEC 25010 as the software quality evaluation model for the developed system.

Review of Related Literature

Scholarship Management in Higher Education

Scholarship programs are important in higher education because they provide financial assistance to deserving students and help improve access to education. However, scholarship management requires the systematic handling of applications, scholarship requirements, eligibility assessments, approvals, monitoring, and reporting. Without an organized system, scholarship processes may become delayed, inconsistent, and difficult to monitor.

Jibrin et al. (2016) explained that manual processing of scholarship applications is time-consuming and error-prone because applicants must submit forms in person, while administrators manually screen, evaluate, and communicate application concerns. Their e-scholarship system allowed students to apply online and receive updates through SMS and email. This supports the need for online application and notification features in scholarship systems.

Ablen et al. (2025) also identified problems in manual scholarship administration, including delayed processing, inaccurate record-keeping, and poor communication among stakeholders. Their SMART-Scholar system used automation, real-time updates, web and mobile interfaces, and large-scale data management to improve scholarship administration. This shows that digital scholarship systems can improve efficiency, accuracy, and transparency.

Digital and Centralized Scholarship Administration

Digital scholarship administration helps reduce manual workload, organize records, and support faster access to information. Centralized systems are especially important when different offices or grantors handle scholarship data. Without integration, administrators may have difficulty identifying duplicate records, monitoring active scholars, and generating reliable reports.

Widyastuti and Prasandha (2025) stated that scholarship administration may face problems such as lack of data integration, delayed disbursement, misallocation, and potential double funding. Their study emphasized that digital systems can support centralized data management, reduce double funding errors, and improve transparency. This is directly related to the developed system, as one of its main goals is to prevent multiple active scholarships through centralized validation.

Al-Ayyubi and Maulana (2023) also highlighted the importance of web-based scholarship systems in improving access to information and rapid dissemination of updates. Their study included features such as a centralized database, search functionality, notifications, application

submission, document management, security, and evaluation processes. These features are similar to the modules planned for the developed system.

Scholarship Recommendation Systems

Scholarship recommendation systems help students identify scholarship opportunities that match their qualifications, needs, and preferences. This is important because students may miss opportunities when scholarship information is scattered or when they are unsure which scholarships fit their profile.

Sulaiman and Ahmad (2023) developed ScholarsAid, a personalized scholarship web application that allows students to search and filter scholarships based on their preferences or needs. Their system used a matching algorithm to provide relevant scholarship options. This supports the developed system's recommendation feature, which help students identify suitable scholarships based on their academic, financial, and profile information.

Daluyon and Bilog (2025) also developed a scholarship management information system using machine learning models. Their study compared classifiers such as Decision Tree, Random Forest, and Naïve Bayes for scholarship recommendation, prescription, and prioritization. The Decision Tree model achieved the highest accuracy in their study, making it a practical basis for the recommendation module of the developed system.

Notification, Tracking, and Reporting

Scholarship systems should not only accept applications but also keep users informed about scholarship updates, application status, requirements, and deadlines. Notifications and reports improve communication between students, grantors, and administrators.

Al-Ayyubi and Maulana (2023) emphasized that web-based scholarship systems can provide real-time updates and notifications about application deadlines, eligibility criteria, and program changes. Jibrin et al. (2016) also included SMS and email notifications in their e-scholarship system to inform applicants about application progress. In the developed system, notifications were used to inform users about scholarship postings, requirements, application status, and monitoring updates.

Reporting is also necessary for transparency and decision-making. Centralized systems can support monitoring and accountability by keeping organized records and audit trails (Widyastuti & Prasandha, 2025). Therefore, the developed system includes dashboards and reports to help administrators and grantors monitor applications, approved scholars, available slots, and scholarship status.

Review of Related Studies and Systems

Development of E-Scholarship System

Jibrin et al. (2016) developed an e-scholarship system to address the limitations of manual scholarship application processing. Their study found that paper-based scholarship processing was time-consuming and error-prone. The system allowed students to apply online and provided notifications through SMS and email. Although the system improved application

submission and communication, it primarily focused on online application processing and did not emphasize scholarship recommendation, duplication prevention, or full scholarship lifecycle monitoring.

Designing a Web-Based Information System for Scholarship Management

Al-Ayyubi and Maulana (2023) designed a web-based scholarship information system to improve access to scholarship information and rapid dissemination of updates. Their system included a centralized database, user-friendly interface, search functionality, notifications, application submission, document management, and security features. While the study supports the development of a web-based scholarship system, it did not focus on preventing duplication through record linkage or on recommendations using a decision tree algorithm.

Scholarship Management Information System Using Machine Learning Models

Daluyon and Bilog (2025) developed a Scholarship Management Information System using machine learning models. Their study used data preprocessing and compared classifiers such as Decision Tree, Random Forest, and Naïve Bayes for scholarship recommendation, prescription, and prioritization. The Decision Tree model achieved the highest accuracy, and the system was evaluated using ISO/IEC 25010. However, the study focused primarily on machine-learning-based scholarship management and did not fully address the prevention of duplicate scholarship records across internal and external records.

ScholarsAid: A Personalized Scholarship Web Application

Sulaiman and Ahmad (2023) developed ScholarsAid, a personalized scholarship web application based on web scraping and matching algorithms. The system helped students search, filter, and receive scholarship recommendations based on their needs and preferences. Although the system supported personalized scholarship discovery, it focused more on scholarship search and recommendation than on administrative approval, monitoring, duplication prevention, and lifecycle management.

SMART-Scholar: A Scholarship Management and Tracking System

Ablen et al. (2025) developed SMART-Scholar, an automated scholarship management and tracking system for Ormoc City. The system addressed delayed processing, inaccurate records, and poor communication by using automation, web and mobile interfaces, real-time updates, and data management. The study is relevant to the developed system because it supports scholarship tracking and transparency. However, it did not focus on algorithm-based duplication prevention or Decision Tree-based scholarship recommendation.

A Smart Scholarship Management System: All-in-One Solution

Amer (2025) developed a smart scholarship management system featuring dashboards, donor management, reporting, role-based access, and automated administrative processes. The system emphasized transparency and accountability. This study supports the use of role-based access, dashboards, and reporting in the developed system. However, some features, such as donor matching, budget tracking, and receipt scanning, are outside the scope of the developed system, as financial disbursement and accounting are not included.

Synthesis of the Reviewed Literature and Studies

The reviewed literature and studies show that digital scholarship management systems help reduce delays, improve data accuracy, enhance communication, and support transparency. Studies on e-scholarship and web-based scholarship systems show that online application, document submission, notifications, centralized databases, and reporting are important features in improving scholarship administration (Jibrin et al., 2016; Al-Ayyubi & Maulana, 2023).

The reviewed studies also show the growing use of intelligent features in scholarship systems. Personalized recommendation systems help students find scholarships suited to their qualifications and needs (Sulaiman & Ahmad, 2023). Machine learning models such as Decision Tree can be used for scholarship recommendation, prescription, and prioritization (Daluyon & Bilog, 2025).

However, most existing studies focus on only one or a few parts of scholarship management. Some systems focus on online applications, some on recommendations, while others focus on tracking or reporting. Few studies combine centralized scholarship management, recommendations, pre-approval duplication prevention, lifecycle monitoring, notifications, and system evaluation in a single centralized platform.

This supported the development of the centralized scholarship management and tracking system for Bulacan State University – Bustos Campus. The developed system combines scholarship posting, application, document submission, scholarship recommendation, pre-approval duplication prevention, approval, monitoring, renewal, notification, and reporting.

The system was also evaluated using TAM for acceptability and ISO/IEC 25010 for software quality.

CHAPTER III:

METHODOLOGY

This chapter presents the research methods and procedures used in developing the centralized scholarship management and tracking system for Bulacan State University – Bustos Campus. This covers the research design, data collection methods, system development process, tools and technologies utilized in the project, and techniques implemented to guarantee the correctness, usability, and reliability of the system.

Research Methods

This study employed a Developmental Research Design to design, develop, and evaluate a centralized scholarship management and tracking system for Bulacan State University – Bustos Campus. Developmental research is appropriate for this study because it focuses on the systematic design, development, implementation, and evaluation of an information system intended to address identified institutional problems.

The developed system was designed to improve the management of scholarship postings, applications, approvals, monitoring, notifications, and reporting processes through a centralized web-based platform. It also incorporates validation mechanisms to help prevent multiple active scholarship allocations and support compliance with institutional scholarship policies.

The research utilized a quantitative approach to evaluate the system's acceptability and software quality. The quantitative approach involved structured survey questionnaires based

on the Technology Acceptance Model (TAM) and the ISO/IEC 25010 Software Quality Model.

Furthermore, the system was developed using the Agile Software Development Methodology. Agile was selected because it supports iterative and flexible system development, allowing continuous improvement of system features throughout the development process. This methodology enabled the researchers to collaborate closely with stakeholders, including students, scholarship grantors, and ISPS coordinators, to gather feedback and refine system functionalities based on actual operational needs.

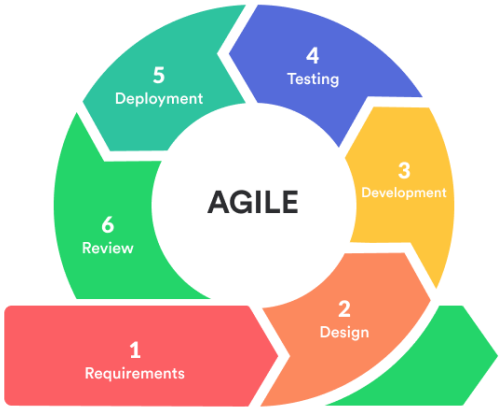


Figure 2. Agile Model

Agile was appropriate for this study because system requirements evolved as stakeholders provided feedback regarding scholarship posting, application, validation, approval, and monitoring processes. Through iterative development cycles, the researchers gradually implemented system modules and presented working prototypes for evaluation. This approach allowed necessary adjustments and enhancements to be incorporated during development rather than at the final stage. By adopting the Agile methodology, the researchers developed a system

that is functional, reliable, and aligned with the operational requirements of scholarship administration at Bulacan State University – Bustos Campus. This ensures that the resulting system remains user-centered, adaptable, and capable of addressing the identified challenges in scholarship management.

Testing

In this phase, the developed system underwent testing procedures to ensure its functionality, reliability, performance, usability, and security. Various testing methods such as functional testing, usability testing, performance testing, and security checking were conducted to identify and resolve system errors before the final evaluation.

Functional testing verified whether each system module performed its expected function, including scholarship posting, student application, required document submission, recommendation, validation, approval, monitoring, notification, and reporting. The Decision Tree recommendation process and the Record Linkage with Levenshtein Distance validation process were also tested using sample student and scholarship records to check whether recommendations and duplication flags are generated correctly. Usability testing determined whether users could navigate and operate the system with ease. Performance testing assessed whether the system can respond to user actions within an acceptable time, while security checking focused on login, authentication, role-based access, and protection of sensitive scholarship records.

The developed system was also evaluated using standardized models, specifically the Technology Acceptance Model (TAM) and the ISO/IEC 25010 Software Quality Model. TAM assessed user acceptance in terms of Perceived Usefulness, Perceived Ease of Use, and Behavioral Intention, while

ISO/IEC 25010 evaluated system quality in areas such as functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

The results of these evaluations determined whether the system meets the expected quality standards and user requirements.

Data Gathering Instrument

This study utilized structured questionnaires as the primary data gathering instrument to evaluate the acceptability and quality of the developed system. The questionnaire was divided into two main sections: Technology Acceptance Model (TAM) and ISO/IEC 25010 Software Quality Model.

The TAM section measured the following:

- Perceived Usefulness
- Perceived Ease of Use
- Behavioral Intention

The ISO/IEC 25010 section measured the following:

- Functional Suitability
- Performance Efficiency
- Compatibility
- Usability

- Reliability
- Security
- Maintainability
- Portability

Sampling Technique

The study employed purposive sampling in selecting the respondents for the evaluation of the Centralized Scholarship Management and Tracking System for Bulacan State University – Bustos Campus. Purposive sampling was used because the respondents were selected based on their direct involvement, experience, or relevant knowledge of the scholarship management process and the functions of the developed system.

The student respondents were selected because they represent the primary users of the system, particularly in viewing scholarship opportunities, submitting applications and documentary requirements, receiving scholarship recommendations, and monitoring application status. Scholarship grantors were selected because they are responsible for managing scholarship programs, evaluating applicants, and processing scholarship decisions. The ISPS Coordinator was included because of the role of the Institutional Student Programs and Services office in managing scholarship records, monitoring scholars, and overseeing scholarship-related processes. IT professionals were selected to evaluate the technical quality of the developed system based on the ISO/IEC 25010 software quality characteristics.

The inclusion criteria for the respondents were as follows: students must be enrolled at Bulacan State University – Bustos Campus and be capable of evaluating the student-related functions of the system; scholarship grantors must be involved in scholarship administration, applicant evaluation, or approval processes; the ISPS Coordinator must be directly involved in the management or monitoring of scholarship-related activities; and IT professionals must possess relevant knowledge or experience in information technology, software development, system evaluation, or related fields.

Through purposive sampling, the researchers ensured that the respondents selected for the study were appropriate to the specific functions and evaluation requirements of the developed system.

The questionnaire was distributed using online platforms such as Google Forms to selected respondents, including students, scholarship grantors, the ISPS coordinator, and IT professionals.

Respondents	No. of Respondents	Description
Students	44	Students of Bulacan State University – Bustos Campus scholarship applications and related requirements.
ISPS Coordinator	1	Personnel responsible for managing scholarship records and processing scholarship requirements and records
Scholarship Grantors	2	Users responsible for managing scholarship programs, evaluating applicants, and approving qualified students within the system.
IT Professionals	5	Experts who evaluate the technical quality and functionality of the system

Total	**52**	Participants involved in system evaluation

Table 1 presents the distribution of respondents involved in the evaluation of the developed centralized scholarship management and tracking system. The respondents include 44 students from Bulacan State University – Bustos Campus who represent the primary users of the system for scholarship application and monitoring. Two scholarship grantors are included because they are responsible for managing scholarship programs, evaluating applicants, and approving qualified students. One ISPS coordinator is included to evaluate administrative and monitoring functions. Five IT professionals assessed the technical quality, functionality, and reliability of the developed system. In total, 52 respondents participated in the evaluation to provide both user-based and technical feedback regarding the acceptability and quality of the developed system.

Data Privacy and Ethical Considerations

The system implements appropriate data privacy and security measures in accordance with the Data Privacy Act of 2012. Access to the system is controlled through role-based authentication to ensure that only authorized users can view or modify scholarship records. Personal data collected from students were used solely for scholarship processing, monitoring, and research evaluation purposes.

The researchers ensured that respondent participation was voluntary and that all responses were treated with confidentiality. Evaluation results were used only for the purpose of assessing the acceptability and quality of the developed system. Personal information,

academic records, and scholarship records were not disclosed to unauthorized individuals.

System Modules

The developed system is composed of several interconnected modules, each designed to perform specific functions that collectively support the complete scholarship management lifecycle. These modules ensure that the processes of posting, application, recommendation, validation, approval, monitoring, notification, and reporting are efficiently handled within a centralized platform. Each module is aligned with the system's objectives and addresses specific challenges identified during the requirements gathering phase.

The system modules are described as follows:

1. Scholarship Posting Module

This module allows scholarship grantors and coordinators to create, update, and manage

scholarship opportunities within the system. It includes functionalities for defining eligibility criteria, application requirements, required documents, deadlines, available slots, scholarship duration, and scholarship descriptions. By centralizing scholarship postings, this module ensures that all available opportunities are organized, easily accessible, and consistently presented to students.

2. Application Module

The application module enabled students to apply for scholarships through an online interface. Students can submit required documents, fill out application forms, and track their application status. This module reduces manual submission processes and ensures that applicant data are

securely stored and readily available for evaluation. Students can only be able to apply to scholarships that are currently posted and available within the system, ensuring that applications are aligned with active scholarship opportunities.

3. Scholarship Recommendation Module

The scholarship recommendation module helps students identify scholarship programs that match their profile and eligibility. It uses student records such as general weighted average, family income, course or program, year level, residency, achievements, and other scholarship-related data and compares them with the eligibility requirements of available scholarship postings.

The module applies a Decision Tree algorithm to classify student eligibility and recommend suitable scholarship opportunities. The Decision Tree approach uses conditional rules to determine whether a student meets specific scholarship criteria. This allows the recommendation process to be explainable because administrators can identify which student attributes influenced the recommendation result.

The basic Decision Tree process includes data collection, data preparation, classification using decision rules, and display of recommended scholarship programs. In this study, the recommendation result serves as a guide for students and administrators, while the final application and approval process still depend on the posted scholarship requirements and authorized evaluation.

4. Scholarship Recommendation Algorithm

The scholarship recommendation process uses a Decision Tree classification approach. Student attributes and scholarship eligibility requirements are used as input variables. The algorithm classifies whether a student is eligible or suitable for a scholarship program based on a series of decision rules.

For classification, the Decision Tree may use information gain to select the attribute that best separates student records according to scholarship suitability. The basic formulas are:

$$Entropy(S) = -\sum_{i=1}^{n} p_i \log_2(p_i)$$

$$Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} \left(\frac{|S_v|}{|S|}\right) Entropy(S_v)$$

The output of the algorithm is a list of scholarship programs that best match the student record. The result supports students in identifying suitable opportunities, but it does not replace the evaluation and approval authority of scholarship grantors or ISPS coordinators.

5. Validation Module

The validation module serves as a core component of the system. It is responsible for supporting the institutional policy of allowing only one active scholarship per student. This module checks existing scholarship records before a new application is approved. To strengthen duplication prevention, the system applies a Record Linkage approach supported by exact matching and Levenshtein Distance for similarity checking. If a conflict or possible

duplicate is detected, the system restricts the approval process or flags the application for review, thereby helping prevent duplicate scholarship allocation before approval.

This proactive validation approach improves data integrity, supports fair distribution of scholarships, and reduces the need for purely manual verification. The validation module covers both internally stored scholarship records and externally sourced scholarship records encoded into the system, ensuring that students cannot receive multiple active scholarships even when scholarships originate from different providers.

6. Duplication Prevention Algorithm

The duplication prevention process begins with exact matching. The system first checks the student identification number against active scholarship records stored in Supabase. If the student identification number already exists in an active scholarship record, the application is flagged and approval is restricted until the authorized user reviews the record.

If the student identification number is missing, inconsistent, or different across records, the system applies Record Linkage with Levenshtein Distance. Record Linkage compares whether two records refer to the same student, while Levenshtein Distance measures the difference between text values such as full names, addresses, or other fields with possible spelling variations.

The system may compute a weighted similarity score using selected fields such as student ID, full name, birthdate, school, course, address, and contact number. The formula is:

$$\text{Score} = \sum_{i=1}^{n} \left(\text{Weight}_i \times \text{Similarity}_i\right)$$

If the computed score reaches the defined threshold, the system marks the record as a possible duplicate. This prevents approval from proceeding automatically and requires administrator or grantor review. This method follows an existing record-matching approach instead of relying only on a newly created rule-based process.

7. Approval Module

The approval module provides scholarship grantors and coordinators with tools to review and evaluate student applications. It allows authorized users to approve, reject, or request corrections from applicants based on eligibility criteria and submitted requirements. Once a student is approved for an active scholarship, the system updates the student's scholarship status and prevents the student from being approved for another active scholarship. The system may also remove or disable the student from other available scholarship applicant lists to support the one-active-scholarship policy. The module ensures a structured and transparent decision-making process while maintaining proper documentation of approvals, rejections, and application updates.

8. Monitoring Module

This module enabled continuous tracking of scholarship records, including scholar status, compliance requirements, academic performance, renewal conditions, and completion. Administrators and coordinators can monitor active scholars, verify compliance, and manage renewal processes efficiently.

9. Notification Module

The notification module provides updates to users regarding application status, missing requirements, deadlines, renewal reminders, monitoring concerns, and important announcements. Notifications are generated and delivered through the system, and email notifications may also be used when applicable. This supports timely communication between students, grantors, and administrators and reduces delays in scholarship processing.

10. Reporting Module

The reporting module generates organized reports and dashboards that summarize scholarship data. These may include statistics on applicants, approved scholars, active scholarships, renewal status, compliance status, and flagged records. The module supports data-driven decision-making by providing administrators and grantors with clear and accessible insights into scholarship operations.

Requirements Documentation

Programming Languages

The system uses JavaScript for the client-side interface and Python for the back-end processes. JavaScript supports interactive web features, while Python handles server-side logic, recommendation processing, validation processes, and communication between the front-end and the database.

Back-End Technology

The back-end of the system was developed using Python. It manages core system operations such as authentication support, application processing, Decision Tree-based scholarship recommendation, Record Linkage and Levenshtein Distance-based validation, approval workflows, monitoring, notification handling, report generation, and communication with the database

Frameworks

The development utilized **React.js** and **FastAPI**. React.js, a JavaScript library, is used to build dynamic and responsive user interfaces for the web application. It supports the creation of role-based dashboards and interactive pages for students, scholarship grantors, ISPS coordinators, and administrators. FastAPI, a Python-based web framework, was used for the back-end to manage application logic, process user requests, implement scholarship recommendation and duplication-prevention functions, and facilitate communication between the front-end and the Supabase database.

Databases

The system uses Supabase as the database platform for storing and managing scholarship records, student information, application data, user credentials, notification records, monitoring records, and system logs. The database structure supports organized relationships between students, scholarships, applications, grantors, and monitoring records to ensure data integrity and efficient retrieval for scholarship management and reporting.

Front-End Technologies

The user interface was developed using the following front-end technologies:

- HTML
- CSS
- JavaScript
- React.js

These technologies allow the development of a responsive and interactive web interface that can be accessed through supported web browsers.

System Architecture

The developed system uses a web-based client-server architecture, which enables centralized data management, real-time access, and efficient communication between users and the system. This architecture is appropriate for the study because it allows multiple users, such as students, scholarship grantors, ISPS coordinators, and administrators, to access the system through standard web browsers without requiring installation on individual devices.

The system is composed of three primary components: the client side or front-end, the server side or back-end, and the centralized database. These components work together to ensure seamless operation, data processing, and secure information management.

1. Client Side (Front-End)

The client side serves as the user interface of the system, where users interact with the platform. It is designed to be accessible through web browsers, allowing users to conveniently access the system using desktops, laptops, or other web-enabled devices. This component provides role-based interfaces tailored for students, scholarship grantors, and administrators. Students

can view available scholarships, submit applications, upload required documents, and track their status, while grantors and administrators can manage scholarship postings, review applications, validate records, monitor scholars, and generate reports. The front-end focuses on usability, accessibility, and responsiveness to ensure a user-friendly experience.

2. Server Side (Back-End)

The server side is responsible for handling the core functionality and logic of the system. It processes user requests, manages system operations, and ensures that transactions are executed accurately and securely. This component implements key functionalities such as application processing, recommendation processing, approval workflows, monitoring, notification handling, reporting, and the pre-approval validation mechanism, which supports the policy of allowing only one active scholarship per student. The back-end also manages communication between the front-end and the Supabase database, ensuring that user inputs are properly processed and that correct information is returned to users.

3. Database (Centralized Storage)

The database serves as the centralized repository of all system data, including student records, scholarship information, application details, required documents, approval history, monitoring records, externally sourced scholarship records, notification records, and transaction logs. It is designed to support updated and consistent information across users. By centralizing data storage, the system minimizes redundancy, reduces inconsistencies, and improves data integrity. The database also plays a critical role in supporting the validation mechanism by storing scholarship records that are checked before approval.

This architecture supports scalability, accessibility, security, and efficient data management for the developed centralized scholarship management and tracking system.

Statistical Treatment of Data

The following statistical formula were used:

Weighted Mean Formula

$$M = \frac{\sum(XW)}{N}$$

Where:

M = Weighted Mean

X = Score

W = Weight

N = Total responses

Standard Deviation Formula

$$\sigma = \sqrt{\frac{\sum(X-\mu)^2}{n}}$$

Where:

s = Sample Standard Deviation

X = Individual Score

$\bar{X}$ = Sample Mean

n = Number of Responses

Σ = Summation of all values

Likert Scale Interpretation

Scale	Range	Descriptive Rating
5	4.21 – 5.00	Very High Acceptability
4	3.41 – 4.20	High Acceptability
3	2.61 – 3.40	Moderate Acceptability
2	1.81 – 2.60	Low Acceptability
1	1.00 – 1.80	Very Low Acceptance

Table 2: presents the Likert scale interpretation used in evaluating the acceptability and quality of the developed system. A score ranging from 4.21 to 5.00 indicates Very High Acceptability, meaning that respondents strongly agree that the system meets the expected quality standards. Scores between 3.41 and 4.20 indicate High Acceptability, showing that the system performs well and satisfies user expectations. A rating of 2.61 to 3.40 represents Moderate Acceptability, suggesting that the system is acceptable but may still require improvements. Scores between 1.81 and 2.60 indicate Low Acceptability, while ratings from 1.00 to 1.80 represent Very Low Acceptance.

This scale were used together with the weighted mean formula to determine the overall acceptability and software quality of the system based on the responses of the participants.

Requirements Analysis

Requirements analysis involves identifying the existing problems in scholarship management at Bulacan State University – Bustos Campus. Through consultations with stakeholders, the study examines current procedures related to scholarship posting, application, validation, approval, monitoring, notification, and reporting.

The analysis reveals several issues, including manual record tracking, difficulty in verifying students with existing scholarships, lack of centralized monitoring, delays in communication between stakeholders, unclear tracking of scholarship status, and the need to define user roles and responsibilities. These findings guided the development of a centralized scholarship management system that manages scholarship records, recommends suitable scholarships, monitors eligibility, and supports the policy of allowing only one active scholarship per student through pre-approval validation.

1. Data Collection and Requirement Analysis

   Relevant scholarship-related data were gathered from the ISPS office, students, and scholarship grantors. These include student information, scholarship records, eligibility data, required documents, and existing monitoring procedures. The collected data guided the design of system functionalities, database structure, and reporting features.

2. User Registration and Login

   The system provides secure authentication with role-based access for students, scholarship grantors, ISPS coordinators, and administrators. Each user type has specific permissions to access system features based on their responsibilities.

3. Scholarship Management and Monitoring

   This includes scholarship posting, application processing, lifecycle management, approval status tracking, renewal monitoring, academic performance monitoring, and scholarship cancellation or termination rules.

4. Scholarship Recommendation Mechanism

   The system provides scholarship recommendations using a Decision Tree approach. Student records are compared with scholarship eligibility requirements to identify suitable scholarship opportunities. The recommendation output guides students in selecting applicable scholarship programs and assists administrators in organizing eligibility-based records.

5. Validation and Approval Mechanism

   The system implements a pre-approval validation mechanism using exact matching and Record Linkage with Levenshtein Distance to check whether a student already has an active scholarship before approval. This prevents duplicate scholarship allocation and supports compliance with institutional policies. Final approval decisions remain under authorized grantors, ISPS coordinators, or administrators.

6. Notification and Reporting System

   The system provides notifications and generates reports and dashboards to support monitoring, communication, and decision-making. Notifications inform users about application status, missing requirements, deadlines, renewal reminders, and monitoring updates.

Development Environment

The system is deployed in a secure web hosting environment that provides centralized access to the developed platform. The deployed environment supports multiple authorized users, centralized data management, and secure access to scholarship-related information. Users can access the system through supported web browsers using compatible desktop, laptop, and other web-enabled devices.

The system interface was designed to support accessibility across desktop and laptop devices and may also be accessed through compatible web-enabled devices. After development and testing, the system was deployed in a secure web hosting environment to ensure reliable access, data security, and centralized scholarship management.

Implementation Plan

The system was developed using the Agile Software Development Methodology, which supports iterative and flexible development. System features were continuously refined based on feedback from stakeholders, including students, scholarship grantors, and ISPS coordinators.

The implementation proceeded through development cycles focusing on core system modules such as scholarship posting, application processing, recommendation, validation, approval, monitoring, notification, and reporting. A centralized database and role-based access control were established to support secure and efficient system operation.

The developed system underwent testing and evaluation using the Technology Acceptance Model (TAM) and ISO/IEC 25010 Software Quality Model. The developed system is currently

deployed and accessible through its web hosting environment. The system continues to support centralized scholarship management while evaluation results and user feedback provide a basis for future improvements, maintenance, and system sustainability.

Implementation Phases

Planning and Designing Phase

In this phase, the researchers finalized the system requirements, system architecture, database structure, user roles, and interface designs. Feedback from stakeholders was considered to ensure that the design reflected the actual scholarship process of Bulacan State University – Bustos Campus.

Development and Prototyping

In this phase, the researchers developed the core modules of the system, including scholarship posting, application, recommendation, validation, approval, monitoring, notification, and reporting. The Decision Tree recommendation process and Record Linkage with Levenshtein Distance validation process were implemented in the back-end. Initial prototypes were presented for review, and necessary improvements were applied based on feedback.

Testing and Feedback

After each development iteration, testing was performed to identify errors, bugs, missing features, or areas for improvement. Feedback from students, scholarship grantors, ISPS coordinators, administrators, and IT professionals was analyzed and used to improve the next development cycle.

Deployment

The developed Centralized Scholarship Management and Tracking System is deployed in a secure web hosting environment and is accessible to authorized users through supported web browsers. The deployment environment supports centralized data management and allows students, scholarship grantors, ISPS coordinators, and administrators to access the system according to their assigned roles.

The system interface was designed to support accessibility across desktop and laptop devices and can also be accessed through compatible web-enabled devices.

The current deployment serves as the operational environment of the developed system and provides the platform used for system access, testing, evaluation, and demonstration.

Post-Deployment Evaluation

The deployed system was evaluated using the Technology Acceptance Model (TAM) and the ISO/IEC 25010 Software Quality Model. The evaluation measured user acceptance and software quality based on the responses of the selected users and IT professionals. The evaluation results and user feedback served as the basis for identifying areas for improvement, future updates, maintenance, and continued development of the system.

CHAPTER IV:

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents, analyzes, and interprets the findings of the study based on the development and evaluation of the **Centralized Scholarship Management and Tracking System for Bulacan State University – Bustos Campus**. It discusses the concerns identified in the existing scholarship management process, presents the developed system and its major functionalities, and evaluates the acceptability and software quality of the system using the Technology Acceptance Model (TAM) and the ISO/IEC 25010 Software Quality Model.

The chapter is divided into three major sections. The first section presents the results of the preliminary study and the identified concerns that served as the basis for system development. The second section presents and describes the major interfaces and functions of the developed system. The third section presents and interprets the results of the system evaluation based on the responses gathered from the selected users and IT professionals.

I. Understanding the Challenges in Scholarship Management

The preliminary study examined the existing scholarship management procedures at Bulacan State University – Bustos Campus. The scholarship process involves activities such as posting scholarship opportunities, receiving student applications, submitting and verifying documentary requirements, evaluating applicants, monitoring approval status, maintaining scholarship records, monitoring scholars, and providing notifications.

The analysis identified that several scholarship-related activities relied on manual or fragmented procedures. This condition may make the processing and monitoring of scholarship records more difficult because information related to applications, requirements, approvals, and active scholarships may be maintained through different sources or processes. The study therefore identified a need for a centralized platform where scholarship-related records and activities could be managed in a more organized manner.

Another concern identified was the monitoring of students who already possess an active scholarship. Scholarship information may originate from internally administered programs as well as externally sourced scholarship records. Without centralized record checking, determining whether a student already has an active scholarship may become difficult and may result in possible duplicate scholarship allocation.

The study also identified the need for improved monitoring throughout the scholarship lifecycle. Aside from the initial application and approval processes, scholarship management includes monitoring academic performance, compliance requirements, renewal conditions, active scholarship status, completion, returning students, and possible cancellation or termination. These activities require updated and accessible records so that authorized personnel can properly monitor scholars.

The identified concerns became the basis for the development of the centralized scholarship management and tracking system. The developed system addresses these concerns through scholarship posting, application processing, scholarship recommendation, pre-approval validation, applicant evaluation and approval, scholar monitoring, renewal monitoring,

notifications, and report generation. These functionalities are intended to provide organized scholarship records and support the different users involved in scholarship administration.

II. Designing and Developing a Centralized Scholarship Management and Tracking System

Based on the identified requirements, the researchers developed a centralized web-based platform that supports the major activities involved in scholarship management.

The **Scholarship Posting Module** enables authorized scholarship grantors and coordinators to create and manage scholarship opportunities. It contains information such as scholarship descriptions, eligibility criteria, documentary requirements, deadlines, available slots, and scholarship duration.

The **Application Module** enables students to apply for available scholarship programs and submit the required information and documents. Students are also provided with a mechanism for monitoring the status of their submitted applications.

The **Scholarship Recommendation Module** compares student information with the eligibility requirements of available scholarship programs. The system uses a Decision Tree approach to identify scholarship opportunities that may correspond to the student's qualifications. The recommendation serves as a guide in identifying possible scholarship opportunities and does not replace the actual evaluation and approval process.

The **Validation Module** performs pre-approval checking of scholarship records. The system initially checks available student and scholarship information and may apply Record Linkage

with Levenshtein Distance to identify similar records containing variations in student information. Records identified as possible duplicates are flagged for review before an application can proceed through the approval process.

The **Approval Module** allows authorized scholarship grantors or coordinators to review applications, submitted requirements, eligibility information, and validation results before processing an application decision. Once an application is approved, the system updates the corresponding scholarship record and active scholarship status.

The **Monitoring Module** maintains information related to active scholars, compliance requirements, academic performance, renewal conditions, completion, returning student status, and scholarship cancellation or termination.

The **Notification Module** provides updates regarding application status, missing requirements, deadlines, renewal reminders, monitoring concerns, and other scholarship-related announcements.

The **Reporting Module** provides organized summaries and reports concerning applicants, approved scholars, active scholarships, renewal status, compliance records, and flagged scholarship records. These modules collectively support the scholarship management lifecycle within the centralized platform.

Presentation and Description of the System

This section presents the major interfaces and functionalities of the developed Centralized Scholarship Management and Tracking System for Bulacan State University – Bustos Campus.

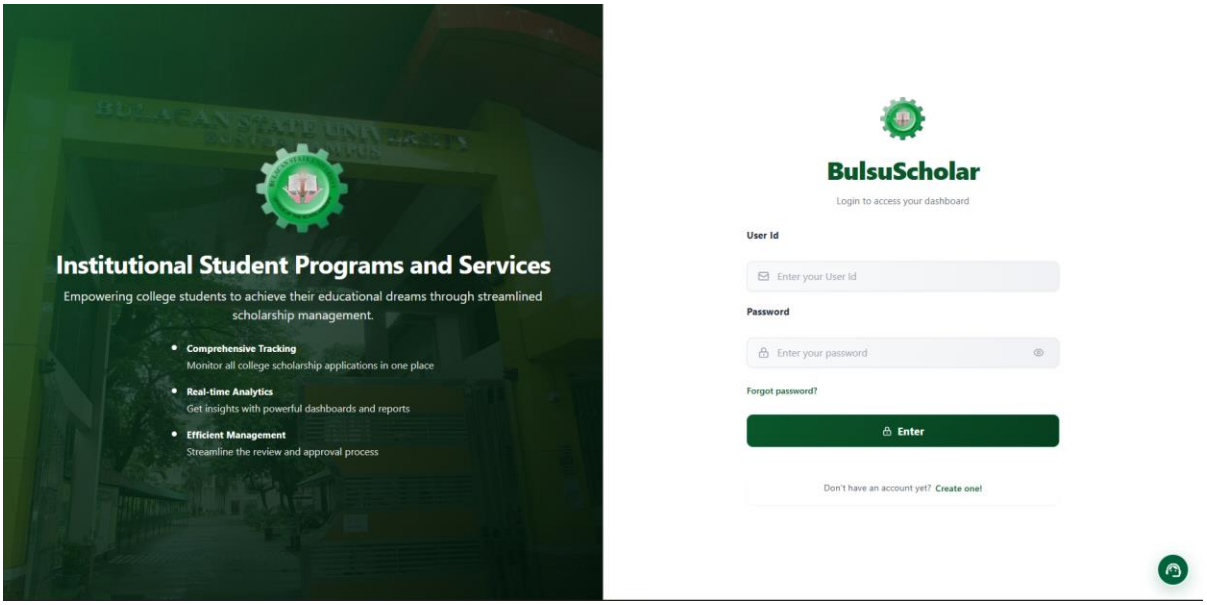


Figure 3. Login Page

Figure 3 presents the login page of the system. The interface allows registered users to enter their authorized account credentials and access the system according to their assigned role. Students, scholarship grantors, ISPS coordinators, and administrators are directed to functions appropriate to their respective responsibilities. The login mechanism also supports role-based access by preventing users from accessing functions that are outside their assigned privileges.

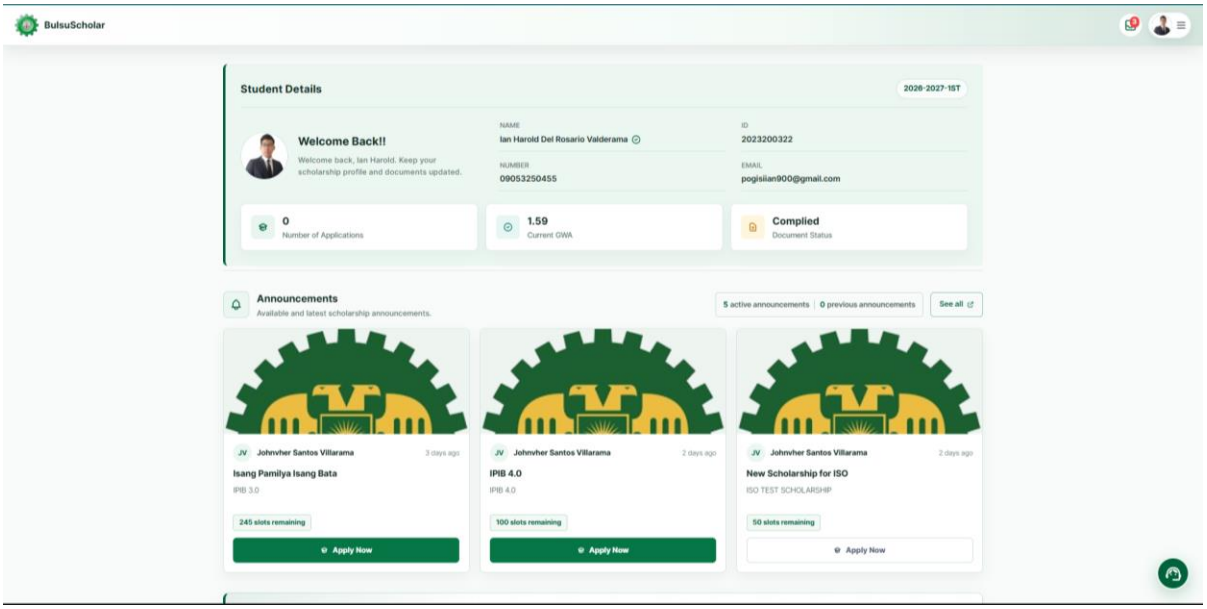


Figure 4. Student Dashboard

Figure 4 presents the student dashboard. The dashboard serves as the main interface for students and provides an overview of scholarship-related information available to them. It gives students access to available scholarship opportunities, submitted applications, current scholarship status, system-generated recommendations, and notifications. By consolidating this information on a single interface, students are able to monitor scholarship activities without relying on separate sources.

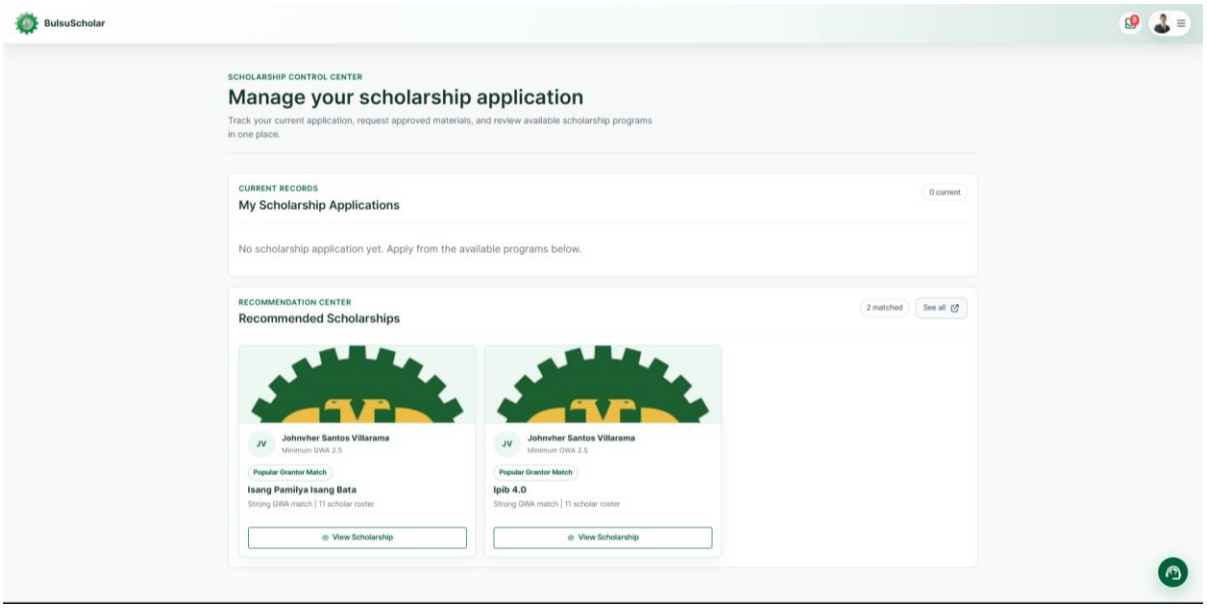


Figure 5. Available Scholarships

Figure 5 presents the available scholarships page. The interface displays scholarship opportunities that are currently posted and available for student application. Relevant information such as the scholarship description, eligibility requirements, documentary requirements, deadline, available slots, and duration is presented to help students determine whether a scholarship is appropriate before proceeding with an application.

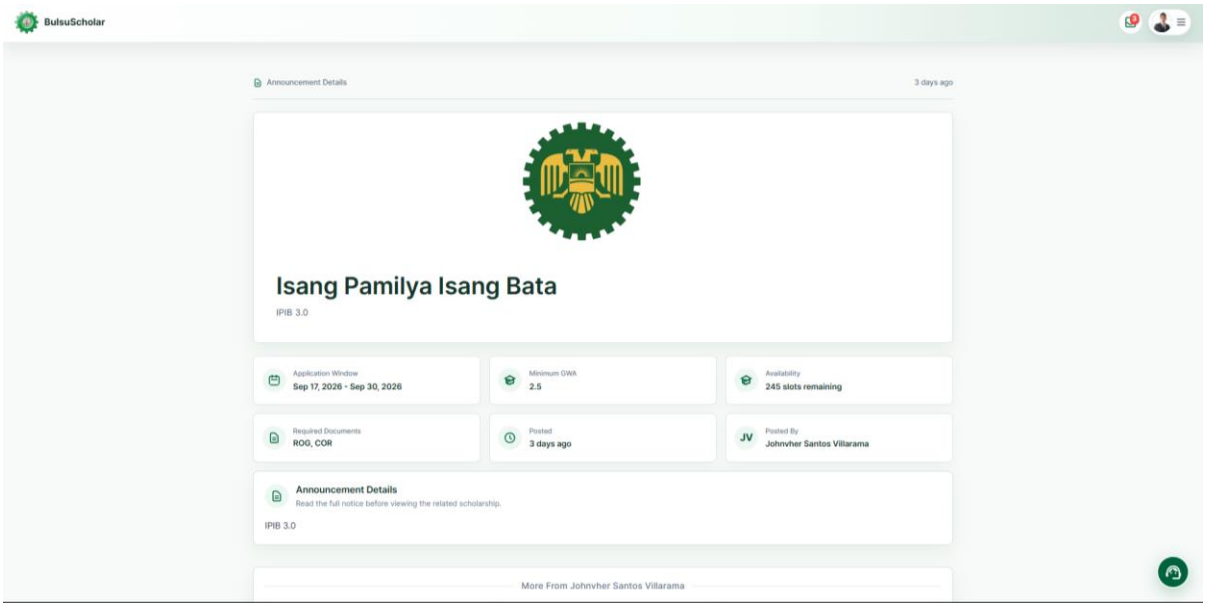


Figure 6. Scholarship Details

Figure 6 presents the detailed information for a selected scholarship program. The page provides students with the complete scholarship information and requirements that must be reviewed before applying. Presenting these details before submission helps ensure that students understand the scholarship conditions and required documents.

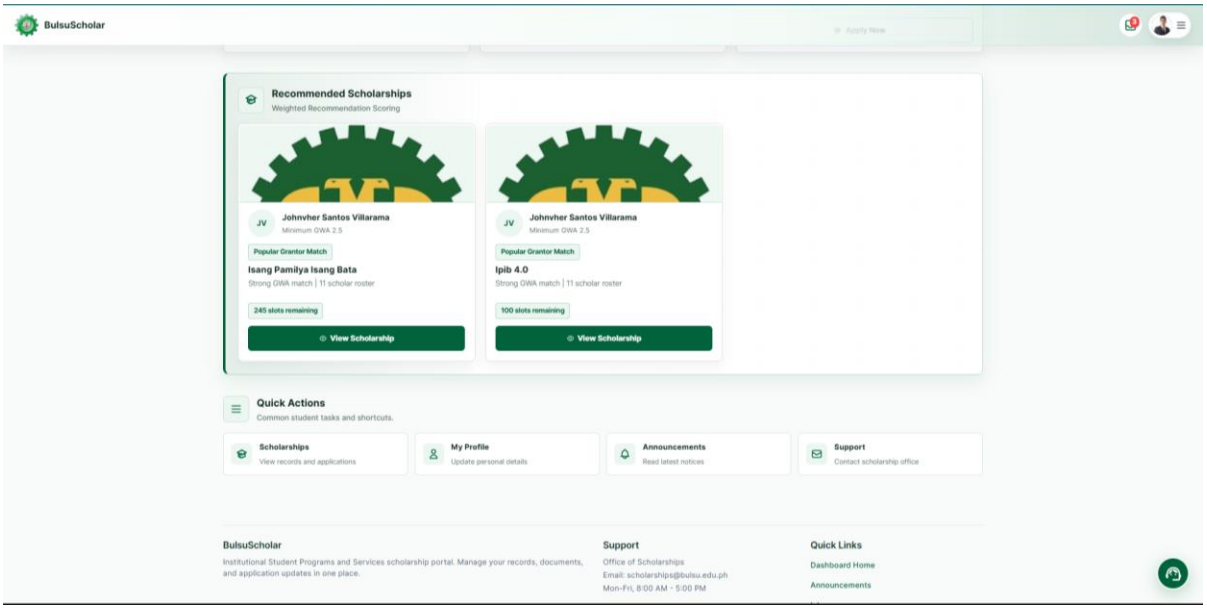


Figure 7. Scholarship Recommendation

Figure 7 presents the scholarship recommendation feature. The system compares the student's available profile and academic information with the eligibility requirements of scholarship programs stored in the system. Based on these criteria, the recommendation process identifies scholarship opportunities that correspond to the student's qualifications. The generated recommendations assist students in identifying potentially suitable scholarships but do not automatically guarantee scholarship approval.

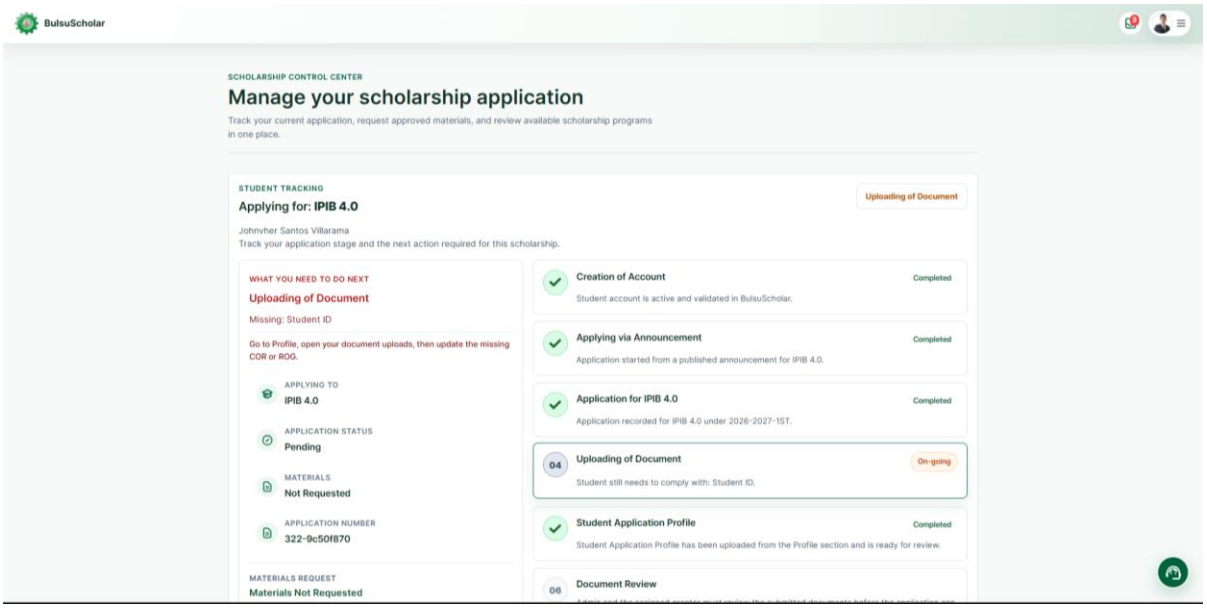


Figure 8. Scholarship Application

Figure 8 presents the scholarship application interface. Students can provide the required application information and submit the necessary documentary requirements for the selected scholarship program. The submitted information is stored in the centralized system and becomes available to authorized scholarship personnel for evaluation.

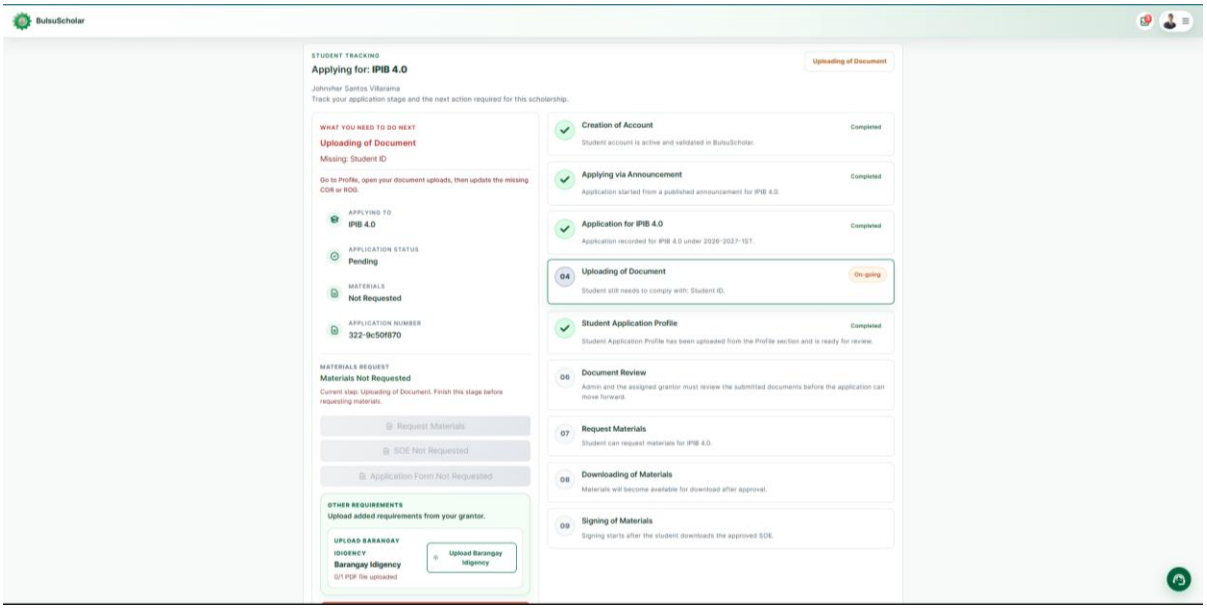


Figure 9. Application Tracking

Figure 9 presents the application tracking feature. The interface enables students to monitor the status of submitted applications and view updates made during the application process. The feature provides greater visibility into whether an application is pending evaluation, requires correction, has been approved, or has been rejected.

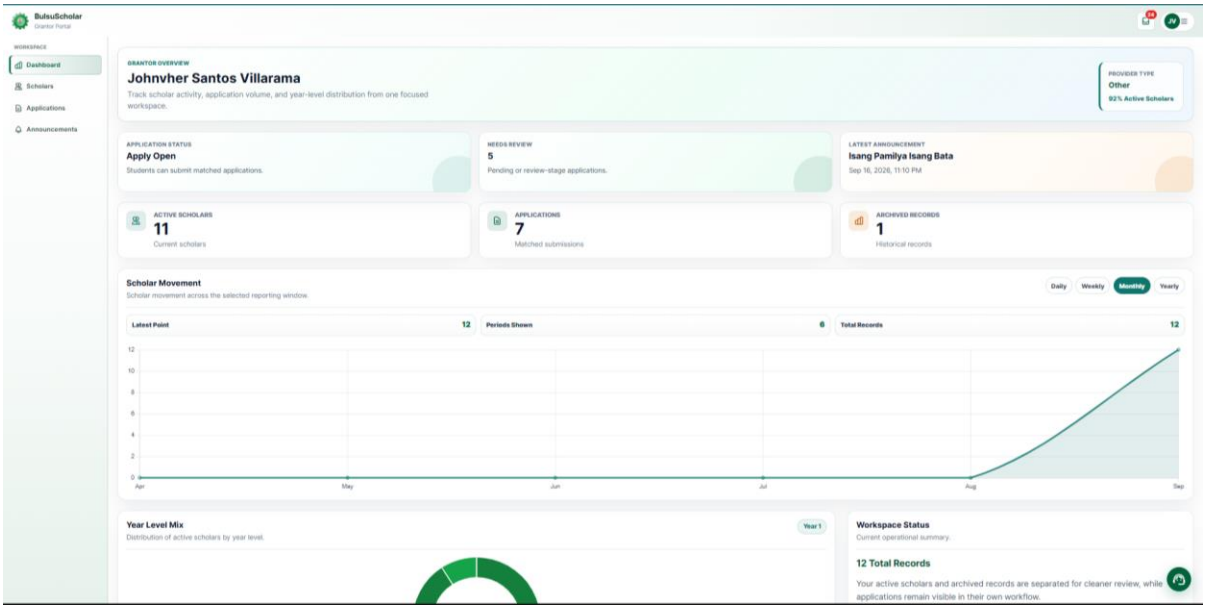


Figure 10. Grantor Dashboard

Figure 10 presents the dashboard provided to scholarship grantors. The dashboard gives authorized grantors access to scholarship postings, applicant records, evaluation functions, approval processes, scholar monitoring information, and other scholarship-related records. It serves as the primary interface for managing the scholarship programs assigned to the grantor.

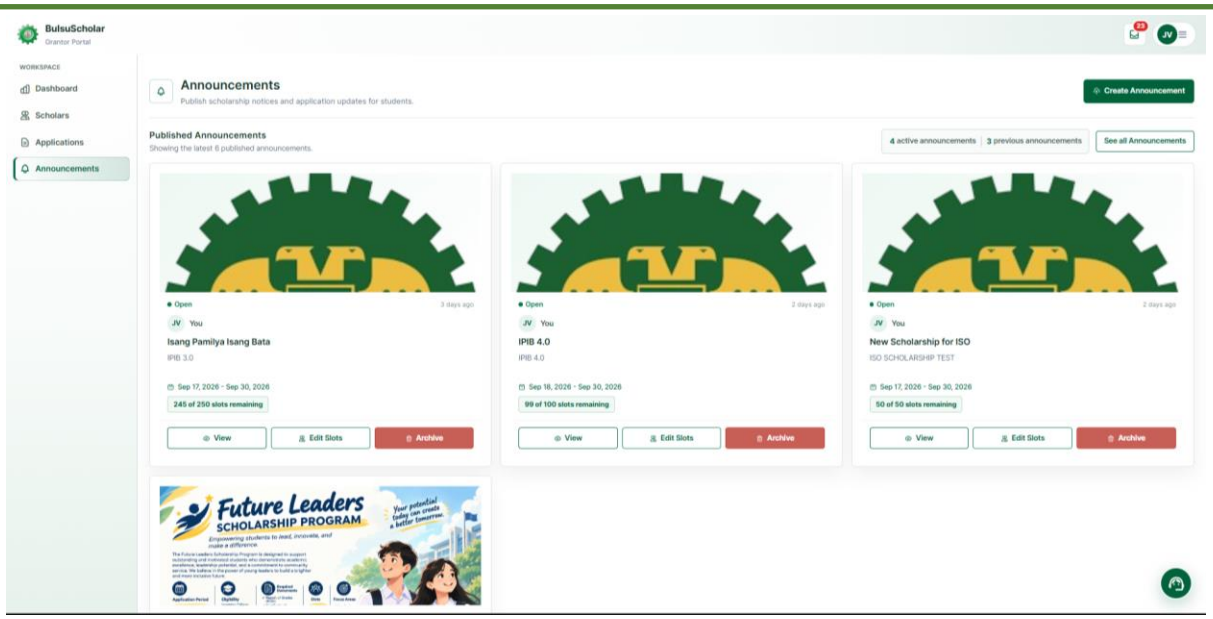


Figure 11. Scholarship Posting

Figure 11 presents the scholarship posting feature. Authorized users can create and manage scholarship opportunities by entering scholarship information, eligibility criteria, documentary requirements, deadlines, scholarship duration, and available slots. Once posted, the scholarship information becomes available to eligible system users.

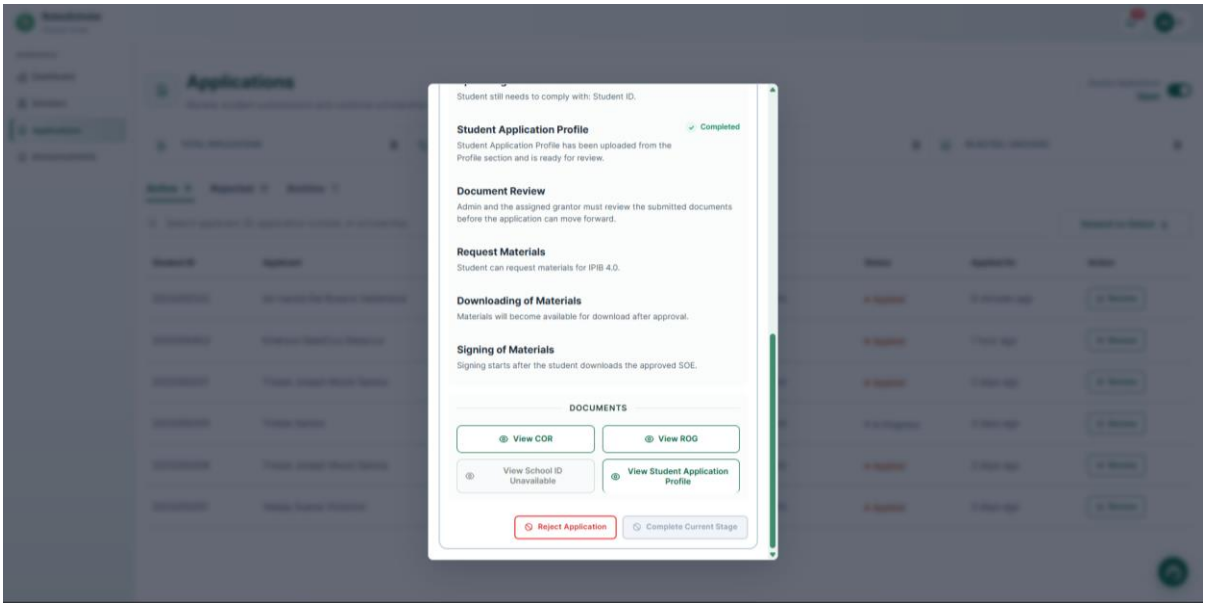


Figure 12. Applicant Evaluation

Figure 12 presents the applicant evaluation interface. Authorized grantors or coordinators can review student information, submitted documentary requirements, and other relevant application information. This allows scholarship personnel to determine whether the applicant satisfies the requirements of the selected scholarship before an approval decision is made.

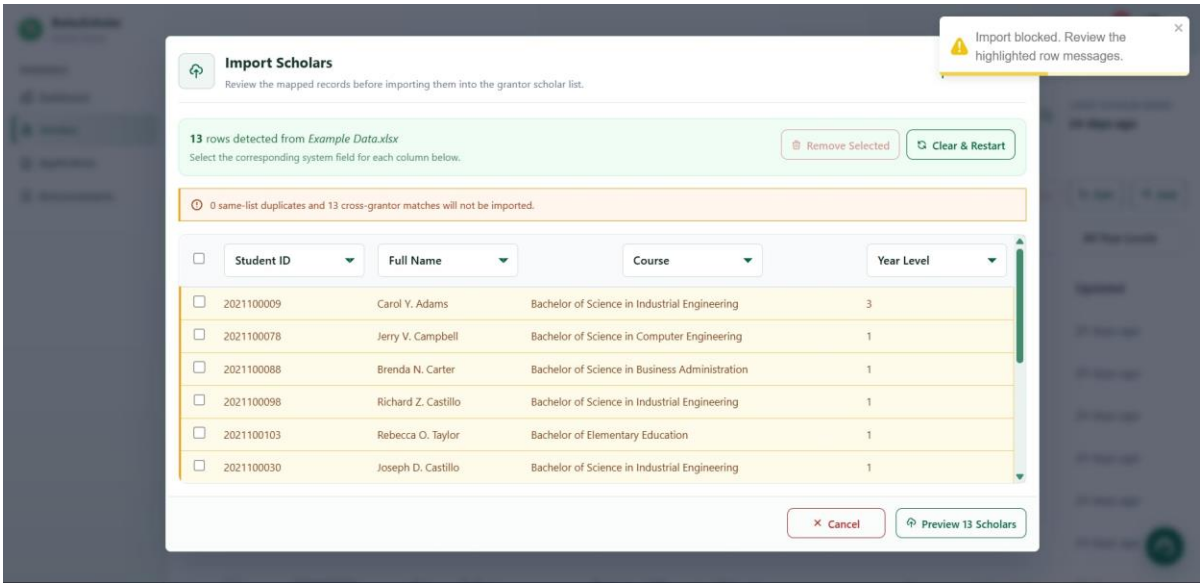


Figure 13. Duplicate Scholarship Validation

Figure 13 presents the duplicate scholarship validation feature. Before an application can be approved, the system checks existing scholarship records to determine whether the student already possesses an active scholarship. The validation process may use exact record matching as well as similarity checking to detect possible variations in student records. When a possible duplicate is identified, the record is flagged for review and the approval process is restricted until the authorized user verifies the result. This feature directly supports the study objective of preventing multiple active scholarships from being granted to the same student.

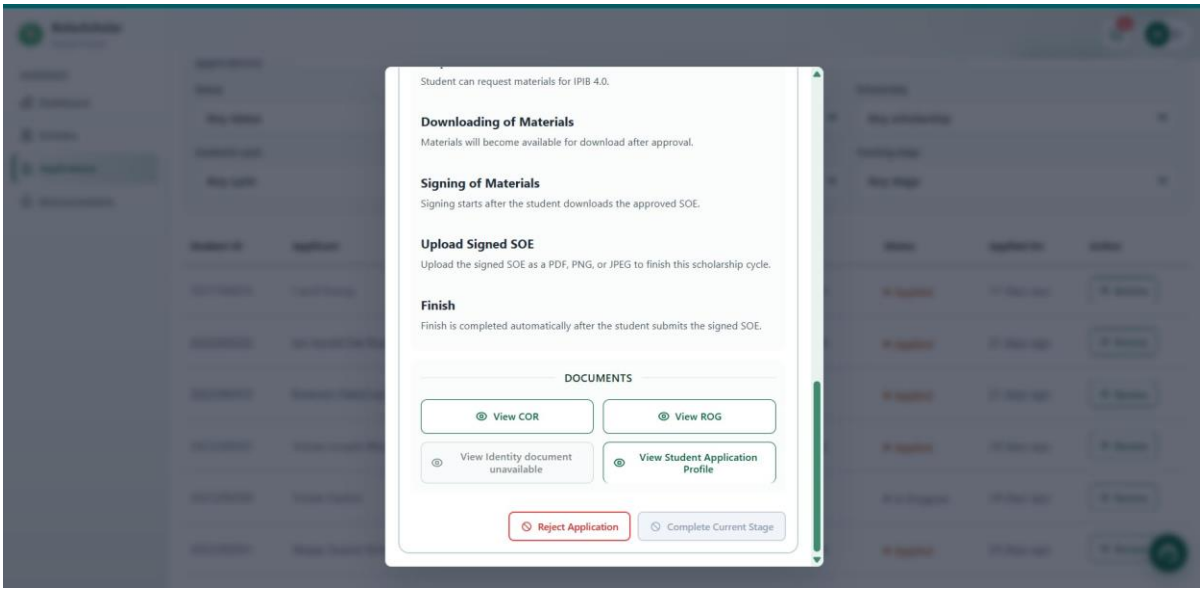


Figure 14. Approval Page

Figure 14 presents the scholarship approval interface. Authorized users can review the application, submitted requirements, eligibility information, and validation result before processing the application decision. Depending on the evaluation, an application may be approved, rejected, or returned for correction. When an application is approved, the system updates the student's scholarship record and identifies the scholarship as active.

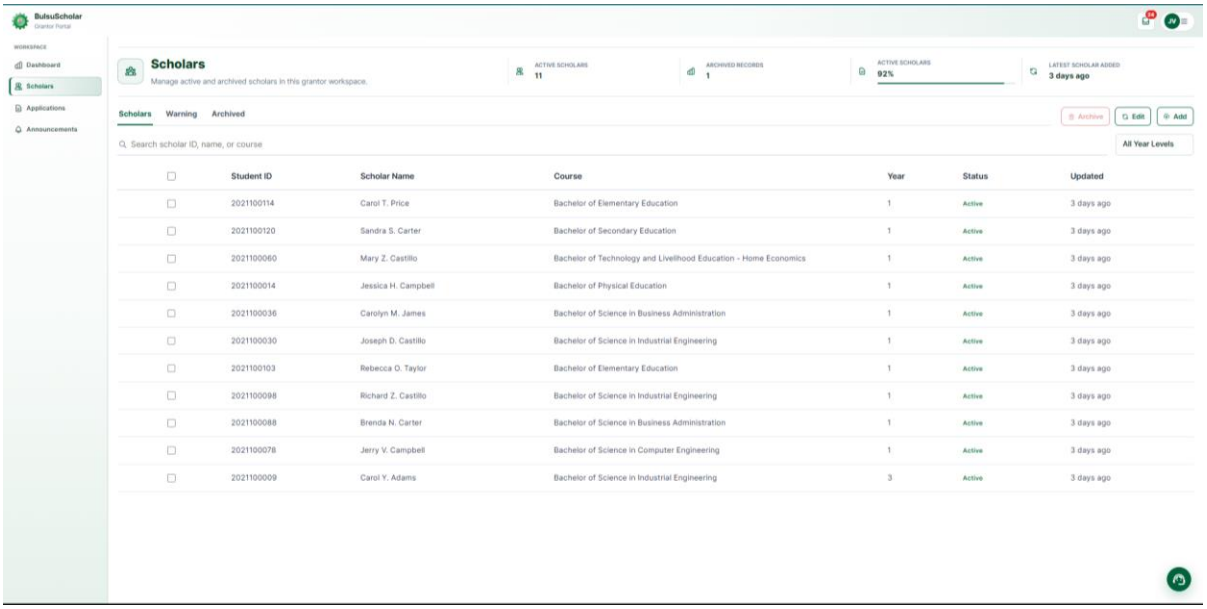


Figure 15. Scholar Monitoring

Figure 15 presents the scholar monitoring interface. The system maintains information concerning active scholars, compliance requirements, academic performance, scholarship status, renewal conditions, completion, returning student status, and other scholarship conditions. The monitoring feature assists authorized users in maintaining updated records throughout the scholarship lifecycle.

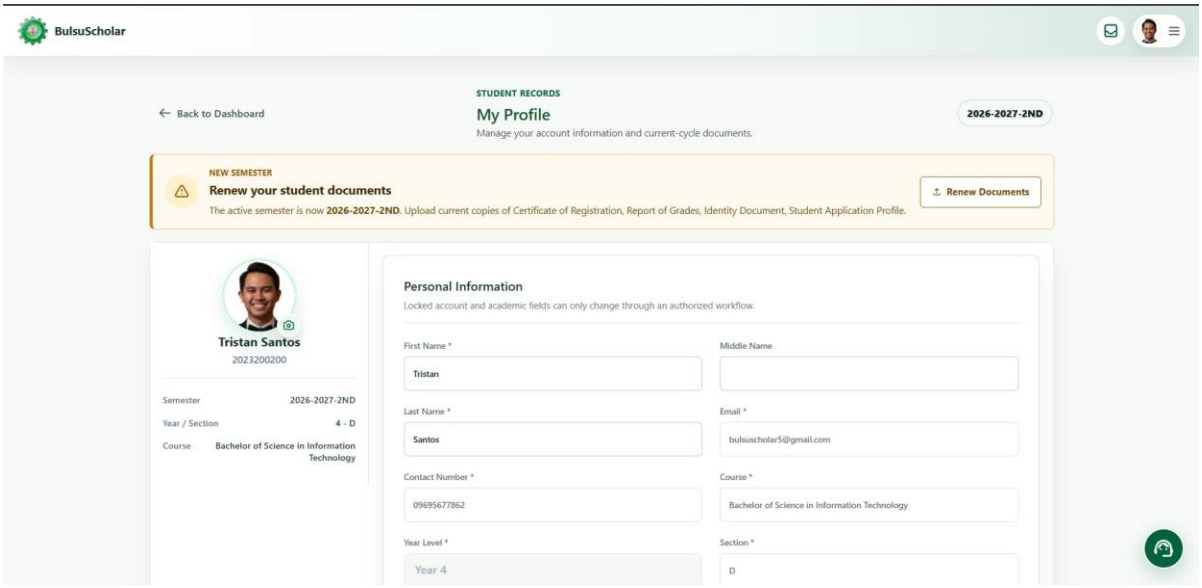


Figure 16. Renewal Monitoring

Figure 16 presents the scholarship renewal monitoring feature. The interface allows authorized users to review renewal requirements and determine the current renewal status of scholars. This functionality supports the continuation of scholarship records beyond the initial approval stage and helps monitor whether scholars satisfy the conditions required for renewal.

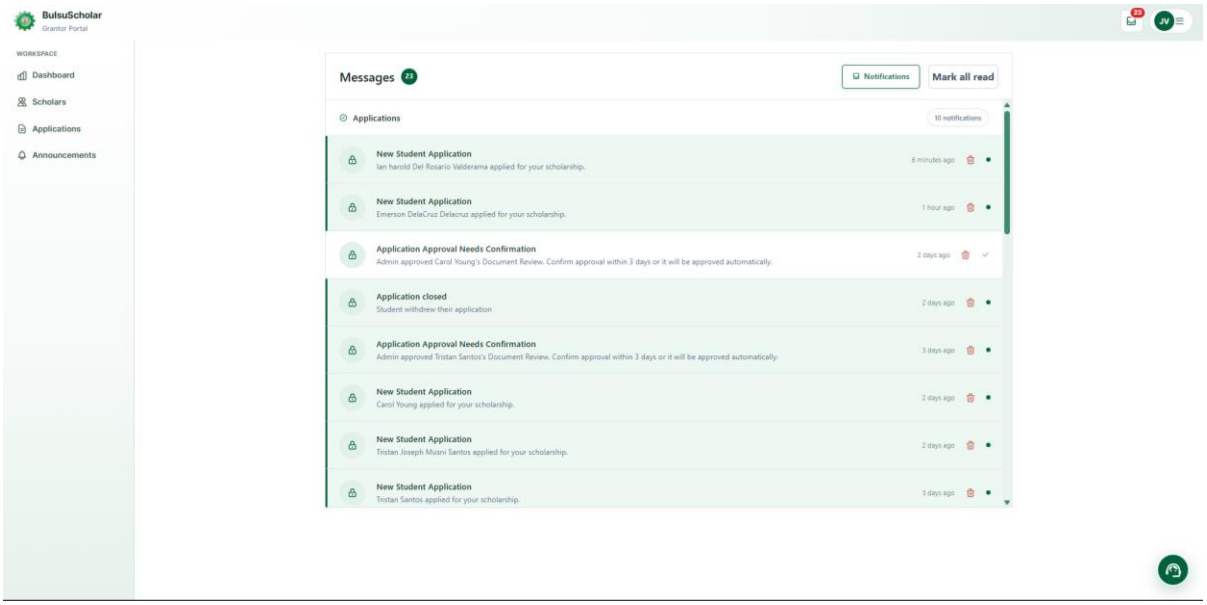


Figure 17. Notifications

Figure 17 presents the notification feature of the system. Notifications provide users with updates concerning application status, documentary requirements, deadlines, scholarship renewals, monitoring concerns, and other important scholarship-related announcements. This feature supports communication between students and scholarship personnel by providing scholarship updates through the centralized platform.

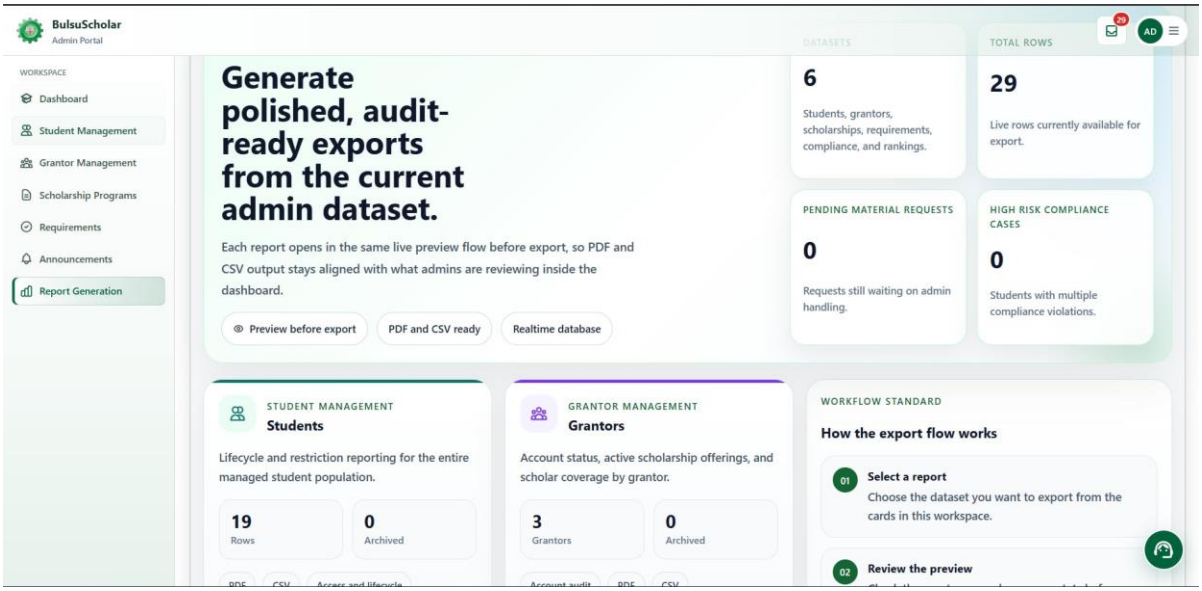


Figure 18. Reports and Dashboard

Figure 18 presents the reporting feature of the system. The system organizes scholarship information into reports and summaries related to applications, approved scholars, active scholarships, renewal status, compliance records, and flagged scholarship records. These reports provide administrators and scholarship personnel with consolidated information that can be used for monitoring and scholarship management activities.

Results and Discussion of the System's Evaluation

This section presents the evaluation of the Centralized Scholarship Management and Tracking System for Bulacan State University – Bustos Campus. The system is evaluated using the Technology Acceptance Model (TAM) and the ISO/IEC 25010 Software Quality Model.

The Technology Acceptance Model evaluates the system in terms of Perceived Usefulness, Perceived Ease of Use, and Behavioral Intention. The ISO/IEC 25010 Software Quality Model evaluates the system in terms of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Users' Technology Acceptance Model (TAM) Evaluation Results

Table 3

Users' Perceived Usefulness Evaluation

Indicators	**WM**	**SD**	***Verbal Interpretation***
The system makes scholarship-related processes faster and more efficient.	4.77	0.43	*Very High Acceptability*
The system helps me access and manage scholarship information more effectively.	4.72	0.58	*Very High Acceptability*
The system makes it easier to monitor scholarship applications and their status.	4.83	0.38	*Very High Acceptability*
The system helps reduce the effort required in performing scholarship-related tasks.	4.81	0.40	*Very High Acceptability*

Overall, the system is useful in improving the scholarship management process.	4.81	0.40	*Very High Acceptability*
Average Mean	**4.79**	**0.33**	***Very High Acceptability***

Table 3 *presents the respondents' evaluation of the system in terms of Perceived Usefulness. The indicators obtained weighted mean scores of 4.77 for faster and more efficient scholarship-related processes, 4.72 for effective access and management of scholarship information, 4.83 for easier monitoring of scholarship applications and their status, 4.81 for reducing the effort required in performing scholarship-related tasks, and 4.81 for the overall usefulness of the system in improving the scholarship management process. The overall average mean of 4.79, with a standard deviation of 0.33, was interpreted as Very High Acceptability. This indicates that the respondents perceived the developed system as highly useful in supporting and improving scholarship-related processes.*

Table 4

Users' Perceived Ease of Use Evaluation

Indicators	**WM**	**SD**	***Verbal Interpretation***
Learning how to use the system is easy for me.	4.55	0.62	*Very High Acceptability*
The system's menus, buttons, and functions are easy to understand.	4.53	0.69	*Very High Acceptability*
It is easy for me to navigate from one part of the system to another.	4.62	0.57	*Very High Acceptability*

I can perform scholarship-related tasks using the system without difficulty.	4.49	0.66	*Very High Acceptability*
Overall, I find the system easy to use.	4.57	0.58	*Very High Acceptability*
Average Mean	**4.55**	**0.53**	***Very High Acceptability***

Table 4*. presents the respondents' evaluation of the system in terms of Perceived Ease of Use. The indicators obtained weighted mean scores of 4.55 for ease of learning the system, 4.53 for the understandability of menus, buttons, and functions, 4.62 for ease of navigation, 4.49 for performing scholarship-related tasks without difficulty, and 4.57 for the overall ease of use of the system. The overall average mean of 4.55, with a standard deviation of 0.53, was interpreted as Very High Acceptability. This indicates that the respondents generally found the system easy to learn, navigate, understand, and operate.*

Table 5

Users' Behavioral Intention Evaluation

Indicators	**WM**	**SD**	***Verbal Interpretation***
I intend to use the system whenever I need to perform scholarship-related activities.	4.68	0.52	*Very High Acceptability*
I am willing to continue using the system in the future.	4.60	0.58	*Very High Acceptability*
I would prefer using this system over a manual scholarship process.	4.83	0.43	*Very High Acceptability*

I would recommend the use of this system to other appropriate users.	4.79	0.41	*Very High Acceptability*
I am willing to use the system regularly when it is implemented.	4.51	0.88	*Very High Acceptability*
Average Mean	**4.68**	**0.41**	***Very High Acceptability***

Table 5: *presents the respondents' evaluation of the system in terms of Behavioral Intention. The indicators obtained weighted mean scores of 4.68 for intention to use the system for scholarship-related activities, 4.60 for willingness to continue using the system in the future, 4.83 for preference for the system over a manual scholarship process, 4.79 for willingness to recommend the system to other appropriate users, and 4.51 for willingness to use the system regularly once implemented. The overall average mean of 4.68, with a standard deviation of 0.41, was interpreted as Very High Acceptability. This indicates a strong intention among the respondents to use, continue using, and recommend the developed system.*

ISO/IEC 25010 Evaluation Results

The system quality evaluation covers the following ISO/IEC 25010 characteristics:

1. Functional Suitability
2. Performance Efficiency
3. Compatibility
4. Usability
5. Reliability
6. Security
7. Maintainability

8. Portability

Separate tables will present the indicators, weighted mean, and verbal interpretation for each software quality characteristic.

Table 6: presents the summary of the IT professionals' evaluation based on the eight ISO/IEC 25010 software quality characteristics.

Indicators	**Average Mean**	**SD**	***Verbal Interpretation***
Functional Suitability	4.40	0.42	*Very High Acceptability*
Performance Efficiency	4.35	0.80	*Very High Acceptability*
Compatibility	4.65	0.42	*Very High Acceptability*
Usability	4.30	0.54	*Very High Acceptability*
Reliability	4.40	0.45	*Very High Acceptability*
Security	4.10	0.76	*High Acceptability*
Maintainability	4.00	0.61	*High Acceptability*
Portability	4.55	0.51	*Very High Acceptability*
Overall Average Mean	**4.34**	**0.44**	***Very High Acceptability***

The eight characteristics and their average mean scores, Functional Suitability **(4.40)**, Performance Efficiency **(4.35)**, Compatibility **(4.65)**, Usability **(4.30)**, Reliability **(4.40)**,

Security **(4.10)**, Maintainability **(4.00)**, and Portability **(4.55)** are presented in Table 6 to show the IT professionals' evaluation of the system. The overall software quality is considered **Very High Acceptability**, as indicated by the overall average mean score of **4.34**.

CHAPTER V:

SUMMARY OF FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

Summary of Findings

After careful analysis of the data gathered and the evaluation results, the summary of findings is presented as follows:

1. The study successfully developed a Centralized Scholarship Management and Tracking System for Bulacan State University - Bustos Campus. The developed web-based system centralizes scholarship posting, student application, document submission, applicant evaluation, approval, scholar monitoring, renewal monitoring, notifications, and report generation within a unified platform for students, scholarship grantors, and administrators.

2. The system implements a pre-approval validation mechanism that checks available scholarship records before an application is approved. Through exact record checking and similarity-based validation, possible duplicate scholarship records can be identified and

flagged for authorized review. This supports the study objective of preventing multiple active scholarships from being granted to the same student.

3. The developed system provides scholarship recommendations and supports scholarship lifecycle management. The recommendation feature assists students in identifying scholarship opportunities based on available profile information and eligibility requirements, while the monitoring functions support the tracking of application, approval, active status, compliance, academic performance, renewal, and scholarship completion.

4. The system was evaluated by 47 user respondents using the Technology Acceptance Model (TAM). Perceived Usefulness obtained an average mean of 4.79, Perceived Ease of Use obtained 4.55, and Behavioral Intention obtained 4.68. All three criteria were interpreted as Very High Acceptability. The overall TAM mean was 4.67, also interpreted as Very High Acceptability, indicating that the respondents generally perceived the system as useful, easy to use, and suitable for continued use in scholarship-related activities.

5. The technical quality of the developed system was evaluated by five IT professionals using the ISO/IEC 25010 Software Quality Model. Functional Suitability obtained 4.40, Performance Efficiency 4.35, Compatibility 4.65, Usability 4.30, Reliability 4.40, Security 4.10, Maintainability 4.00, and Portability 4.55. The overall average mean was 4.34, interpreted as Very High Acceptability. Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, and Portability received Very High Acceptability, while Security and Maintainability received High Acceptability.

Conclusions

After the system was developed, demonstrated, tested, and evaluated, the researchers concluded that the major objectives of the study entitled "Centralized Scholarship Management and Tracking System for Bulacan State University - Bustos Campus" were achieved.

The developed web-based system provides a centralized platform where students can view scholarship opportunities, submit applications and required documents, receive scholarship recommendations, and monitor the status of their applications. Scholarship grantors and administrators are provided with functions for scholarship posting, applicant evaluation, approval, scholar monitoring, renewal monitoring, notifications, and report generation.

The pre-approval validation mechanism supports the prevention of duplicate scholarship allocation by checking available scholarship records before an application is approved. Possible duplicate records can be identified and flagged for authorized review, helping support the institutional policy of allowing only one active scholarship per student.

The scholarship recommendation and monitoring functions also support the different stages of the scholarship lifecycle. Recommendations assist students in identifying potentially suitable scholarship opportunities, while monitoring functions allow authorized users to keep track of active scholarships, academic performance, compliance requirements, renewal status, and scholarship completion.

Based on the final Technology Acceptance Model evaluation, the system obtained an overall mean of 4.67, interpreted as Very High Acceptability. Perceived Usefulness obtained 4.79, Perceived Ease of Use obtained 4.55, and Behavioral Intention obtained 4.68, with all three

criteria interpreted as Very High Acceptability. These results indicate that the intended users generally perceived the system as useful, easy to use, and appropriate for continued use.

The ISO/IEC 25010 evaluation conducted by IT professionals resulted in an overall average mean of 4.34, interpreted as Very High Acceptability. The results indicate that the developed system generally met the assessed software quality characteristics, while Security and Maintainability, which received High Acceptability ratings, remain areas that may be further strengthened.

Overall, the findings indicate that the developed system provides an acceptable and organized centralized approach to scholarship management and tracking at Bulacan State University - Bustos Campus. By consolidating scholarship records, applications, validation, monitoring, notifications, and reporting within one platform, the system supports more efficient scholarship administration and improved access to scholarship-related information according to the users' respective roles.

Recommendations

The researchers recommend that future researchers and developers continue improving the Centralized Scholarship Management and Tracking System by extending its features and addressing the current limitations identified during the development of the study.

The researchers recommend improving student-record verification, scholarship validation, and duplicate detection within the system. Since the accuracy of scholarship recommendations, validation results, monitoring records, reports, and duplicate scholarship prevention depends on the correctness and completeness of the information stored in the

database, future developers may strengthen the validation process by using additional verification fields, improved record-matching methods, and stricter checking of incomplete, inconsistent, outdated, or incorrectly encoded data. These improvements may help reduce false matches, missed duplicate records, and inaccurate validation results.

The researchers also recommend the future integration of external scholarship databases or providers. The current system can include externally sourced scholarship records for monitoring and duplicate checking, but it does not provide full real-time integration with external scholarship providers. Future development may consider API-based integration with government agencies, private scholarship providers, or other authorized scholarship databases, subject to appropriate access permissions, data-sharing agreements, and security requirements. This may improve the timeliness, completeness, and reliability of scholarship records used for validation and monitoring.

Future researchers may also consider developing a dedicated mobile application for the system. The current system is a web-based platform that can be accessed using supported web browsers. A mobile application may provide students, scholarship grantors, and administrators with another convenient means of accessing scholarship notifications, application information, monitoring records, and other scholarship-related services.

Lastly, future researchers may consider expanding the Centralized Scholarship Management and Tracking System for use on a larger scale beyond Bulacan State University – Bustos Campus. Future development may extend the system to other Bulacan State University campuses, such as the Malolos Campus and San Rafael Campus, while adapting the system

to their respective scholarship processes, policies, user requirements, and administrative workflows. Expanding the system across multiple campuses may provide a more unified approach to scholarship management and allow scholarship records, monitoring, validation, and reporting to be handled more consistently across the university.

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